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Rhizome Neural Operator

Technical Report: a state-dependent integral operator for 21 cm reionization fields

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Abstract

We study a recurrent neural integral operator for predicting neutral-fraction and 21 cm brightness-temperature fields from `21cmFAST` lightcones. A learned convolution is multiplied by channel-wise source and receiver gates computed from the current hidden field at each update. This factorization makes the effective interaction depend on the field while retaining FFT evaluation and $\mathcal{O}(N \log N)$ spatial cost at fixed width and mode count. We describe its continuum form, discrete evaluation and restrictions, including the distinction between bounded hidden states and convergence. In the reported single-seed experiments, the 3-D rhizome reaches a neutral-fraction test RMSE of 0.059 against 0.068 for the existing FNO-based pipeline; with joint prediction, the errors are 0.050 and 5.8 mK against 0.056 and 6.5 mK. A slice-wise 2-D model with ±32 cells of LOS context improves morphology further. The best multi-field model, trained on a larger set of cones, reduces bubble-size error from 1.21 to 0.56 Mpc and front width from 6.17 to 3.89 Mpc, and predicts a full cone in 5.5 s on an A100. Adding a brightness-temperature power loss suppresses excess small-scale power before reionization at a 4–8% cost in per-voxel RMSE. These results support the complete prediction pipeline; the separate contribution of evolving-state gates remains to be established by controlled ablations.

Keywords: neural operators, Fourier neural operator, operator learning, 21 cm cosmology, epoch of reionization, emulation

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1 NEURAL OPERATORS AND FOURIER LAYERS

1.1 Operator learning: problem setup

A neural operator learns a map between fields rather than between vectors of a prescribed size. Let $\mathcal{A}$ and $\mathcal{U}$ be normed spaces of functions on domains D and D', and let $\mathcal{G} : \mathcal{A} \to \mathcal{U}$ denote the target map. Given input–output pairs (f_j, g_j), with $g_j = \mathcal{G}(f_j)$, we seek an approximation $\mathcal{G}_\theta$ whose parameters can be used on different discretizations of those fields [5]. The domains need not be identical, although the architectures considered here predict on the input grid.

In this work, the input contains the matter-density field and, for the multi-field models, the LOS velocity. Redshift and simulation parameters specify which map is being emulated; they are conditioning variables, distinct from the learned weights θ. The outputs are the neutral hydrogen fraction x_{HI} and, where included, the brightness temperature T_b. The target is the map defined by the simulation and its preprocessing, rather than an exact solution operator of a specified PDE.

The function-space viewpoint separates an architecture from its numerical evaluation. It does not by itself guarantee continuity, resolution transfer or generalization. For example, L^2 is useful for field-error norms, whereas bounded measurable fields in L^∞ are convenient for the gated interaction below. Point samples require a sufficiently regular representative; simulation cells can instead be understood

as cell averages. Changing the grid must preserve the domain, physical units and meaning of every input channel.

1.2 Neural operators as stacked kernel integrals

A linear integral operator maps a field f to a field g through a kernel:

$$g(r) = \int_D \kappa(r, r'; \theta) f(r')\, \mathrm{d}r'. \tag{1}$$

For vector-valued fields, κ is a matrix that mixes channels. A Green's function gives one physical example of such a kernel, when the inverse of a linear differential operator admits an integral representation. A learned kernel need not be a Green's function or satisfy a differential equation.

An integral alone is linear when the kernel is fixed. A neural operator obtains a nonlinear map by combining integral interactions with pointwise transformations. A typical hidden layer is

$$v_{\ell+1}(r) = \sigma(W_\ell v_\ell(r) + b_\ell(r) + (\mathcal{K}_\ell v_\ell)(r)), \tag{2}$$

with lifting $v_0 = P(f)$ and projection $\mathcal{G}_\theta(f) = Q(v_L)$. Here W_ℓ mixes channels at one position, while $\mathcal{K}_\ell$ exchanges information between positions. This distinction matters: a fixed kernel does not make the complete network linear or its response independent of the input.

1.3 The Fourier neural operator

On a periodic domain, a translation-invariant kernel depends only on separation, $\kappa(r, r'; \theta) = \kappa(r - r'; \theta)$. Its integral is a convolution:

$$\mathcal{K} f = \mathcal{F}^{-1}\left[\hat{\kappa}\, \mathcal{F}[f]\right]. \tag{3}$$

The Fourier neural operator (FNO) parameterizes the matrix-valued multiplier $\hat{\kappa}$ directly [7]. A lifting layer maps the input into d_v channels, several layers of the form Equation (2) combine spectral and pointwise paths, and a projection produces the output (Figure 1).

For a uniform grid, the integral is approximated by

$$g(r_i) = \sum_{j=1}^{N} \kappa(r_i - r_j; \theta) f(r_j) w_j, \tag{4}$$

where w_j is a cell-volume weight, not a one-dimensional spacing unless $d = 1$. On the unit-volume domain used below, $w_j = 1/N$. The FFT evaluates this discrete convolution without forming all N^2 pairs.

1.4 Discretization invariance and its limits

Each spectral path retains a finite set of Fourier indices Λ:

$$\widehat{\mathcal{K} f}(k) = \begin{cases} R_\phi(k) \hat{f}(k), & k \in \Lambda, \\ 0, & k \notin \Lambda, \end{cases} \tag{5}$$

with $R_\phi(k) \in \mathbb{C}^{d_v \times d_v}$. We use $k_{\max}$ for the mode-count setting and Λ for the actual retained set. The implementation uses rectangular, possibly anisotropic bands rather than a spherical cutoff. On a box of side L, index k corresponds to physical angular wavenumber $2\pi k/L$; a mode count is therefore not itself a wavenumber in Mpc^{-1}.

The number of spectral parameters depends on $|\Lambda|$ and d_v, rather than on the grid size. Lifting, projection and pointwise maps also share their weights across cells. The weights can therefore be reused on finer grids of the same domain that resolve the retained modes. This is the architectural basis of the FNO's demonstrated zero-shot super-resolution [7]. Several qualifications matter here.

The spectral path is band-limited; the output need not be. The pointwise path can retain high-frequency information already present in the hidden field, and nonlinear activations can generate additional frequencies. It is therefore incorrect to describe the complete FNO output as band-limited. A finer grid provides additional evaluation points, but reliable predictions on newly resolved physical scales still depend on what the model learned and on the information in its inputs.

Aliasing. Nonlinear activations and products can create frequencies above the grid's Nyquist limit. Sampling folds these frequencies into the represented band, with an error that depends on the grid. Sharing parameters across resolutions is weaker than requiring the discrete evaluations to represent exactly the same continuous operator. This distinction is formalized by Bartolucci et al. [2].

Resolution-dependent targets. Changing the simulation resolution may change smoothing, sub-grid prescriptions and the resolved ionization fronts. Transferring between discretizations of one field and predicting a new simulation at finer resolution are different tests. An architecture that supports the former need not succeed at the latter.

Boundary conditions. The FFT evaluates a periodic convolution. This matches coeval periodic boxes and the transverse directions of the lightcones. Along the non-periodic LOS, padding and cropping define a different bounded-domain operation. Padding reduces wrap-around effects only over the separations it actually separates; it does not impose a physical boundary condition by itself.

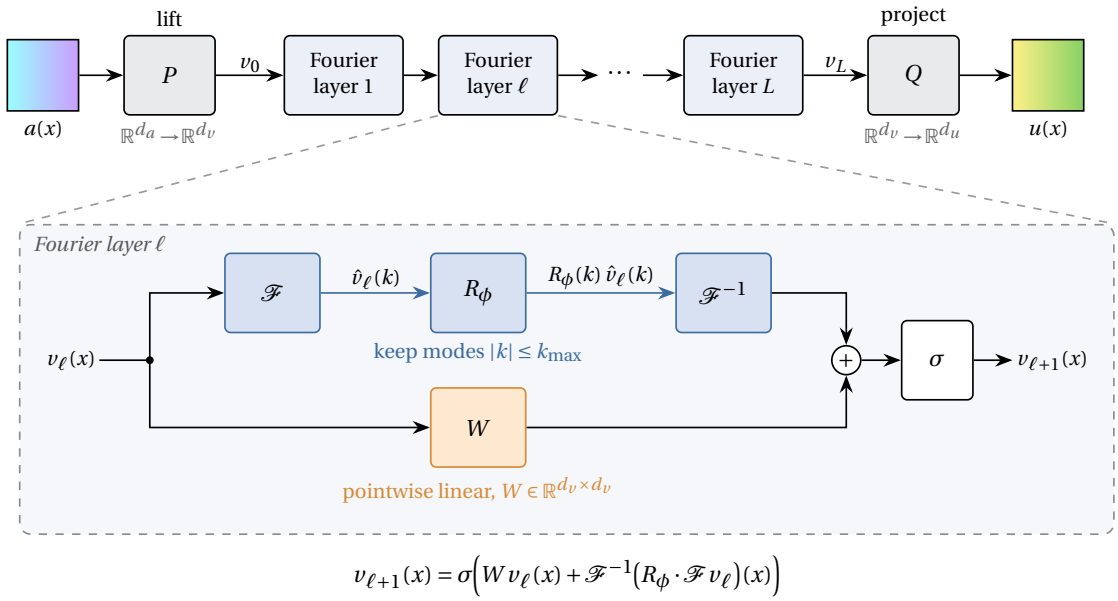
1.5 Limitations of a stationary spectral interaction

The FNO offers an efficient global interaction. For ionization fields, the questions are whether its retained modes and pointwise nonlinearities can represent sharp boundaries and whether the learned response adapts sufficiently to heterogeneous regions.

Stationarity of the linear kernel. A spectral path applies the same linear kernel at every position. The full network can still have an input-dependent response: nonlinearities, conditioning and repeated layers change its Jacobian. What a standard spectral path lacks is an explicit source–receiver modulation of its kernel within that interaction. This is the distinction addressed by the rhizome.

Sharp features. A discontinuity has a broad Fourier spectrum; for example, the coefficients of a one-dimensional step decay as $1/|k|$. Truncating a linear Fourier representation can smooth the edge and produce Gibbs ringing. Ionization maps are close to binary, so front placement is sensitive to these scales. Pointwise paths and activations can recover detail, but a small spectral band does not directly transport arbitrary high-frequency structure between cells.

Intermediate scales. A standard FNO has no explicit spatial hierarchy or compact local stencil. It can represent



$$v_{\ell+1}(x) = \sigma\Big(W v_\ell(x) + \mathcal{F}^{-1}\big(R_\phi \cdot \mathcal{F} v_\ell\big)(x)\Big)$$

Figure 1. Fourier neural operator architecture. The spectral path is linear; the pointwise activation makes the complete layer nonlinear.

interactions at intermediate distances through its kernel, so these scales are not absent; they simply have no dedicated branch as they do in a U-Net.

Cost of more modes. A dense multiplier stores d_v^2 complex entries per retained mode. Its parameter count grows as $\mathcal{O}(|\Lambda| d_v^2)$, with $|\Lambda|$ growing cubically with an isotropic mode count in three dimensions. Factorizing the channel matrices can reduce this cost, but adds a restriction of its own.

These observations motivate a different interaction law; they do not establish an impossibility result for FNOs. The experiments must determine whether the proposed change improves front placement, cross-correlation and morphology under comparable inputs and training conditions.

2 SPATIAL STRUCTURE IN NEURAL OPERATORS AND THE U-NET

Ionized bubbles span a wide range of sizes, while their boundaries can be sharp on the grid scale. This motivates three useful architectural choices: local interactions, a hierarchy of scales, and explicit modulation by the field. Existing models address these in different ways. This section reviews the U-Net and U-FNO before introducing the particular combination used by the rhizome.

2.1 Inductive biases: locality, scale and stationarity

The spatial interaction in a neural-operator layer can often be written as a kernel integral, with pointwise maps around it. Restrictions on this interaction determine some of the architecture's inductive biases. Three properties are useful for comparison.

Receptive field. A convolutional neural network (CNN) uses kernels with compact support of a few grid cells, $\kappa(r - r') = 0$ for $|r - r'|$ larger than the stencil. A single layer is therefore strictly local; the receptive field grows only linearly with depth. The FNO is the opposite extreme: every Fourier layer couples every point to every other point.

Scale. A plain CNN or FNO processes the field at one resolution. Neither has a built-in notion of coarse and fine scales; scale-dependent behaviour has to be learned implicitly. Multi-scale architectures instead process the field at several resolutions and combine the results, so that large-scale context and small-scale detail are handled by separate parts of the network.

Stationarity and state dependence. Convolutional filters and standard Fourier multipliers are fixed after training. Their surrounding nonlinearities still make the complete network's response depend on the input. Attention [11] and graph kernel networks [6] put feature dependence directly into the interaction weights. Dense pairwise evaluation costs $\mathcal{O}(N^2)$ at fixed width, but this is not a lower bound for state dependence: sparse graph interactions and factorized or linear attention reduce that cost [4]. The rhizome uses a convolution with separable endpoint gates.

Table 1 summarizes these properties for the architectures discussed in this report.

2.2 The U-Net

The U-Net [10] was introduced for biomedical image segmentation, a task that, like predicting an ionization field, requires sharp, pixel-accurate boundaries informed by large-scale context. It consists of three parts (Figure 2).

Contracting path. The encoder applies a stack of small convolutions followed by a nonlinearity, then halves the resolution by pooling, and repeats. Denoting the feature map at level s by e_s,

$$e_0 = C_0(a), \qquad e_s = C_s\big(\mathrm{Down}(e_{s-1})\big), \quad s = 1, \dots, S, \tag{6}$$

where each C_s is a block of 3×3 convolutions with nonlinearities. The number of channels is typically doubled at each level, so the network trades spatial resolution for feature richness. Because every level operates on a grid that is coarser by a factor of two, a 3×3 convolution at level s spans $3 \cdot 2^s$ cells of the original grid. The receptive field therefore

Table 1. Spatial interactions in the architectures discussed here. "Fixed" refers to the learned linear filters, not to the full nonlinear response. Costs hold at fixed width; pooling and boundaries can break exact translation equivariance.

Architecture	Receptive field	Base filter	Scale hierarchy	Explicit gates
CNN	local, grows with depth	fixed	no	no
FNO	global spectral path	fixed	no	no
U-Net	local → wide via pooling	fixed	yes	no
U-FNO	global + multi-scale local	fixed	yes	no
Dense attention	global	adaptive	no	$\mathcal{O}(N^2)$
Rhizome	global	fixed + gates	no	$\mathcal{O}(N \log N)$

grows exponentially with the number of levels instead of linearly with depth.

Bottleneck. At the coarsest level, e_S, the field is represented on a grid of $n/2^S$ points per dimension, where n is the original side length with many channels. Here a few convolutions see a large fraction of the domain and capture large-scale context.

Expanding path and skip connections. The decoder upsamples step by step and, at each level, concatenates the upsampled features with the encoder features of the same resolution:

$$d_S = e_S, \qquad d_s = C'_s\big([\,\mathrm{Up}(d_{s+1}),\, e_s\,]\big), \quad s = S-1,\dots,0, \tag{7}$$

and a final 1×1 convolution maps d_0 to the output u. The skip connections $e_s \to d_s$ are the key ingredient. Pooling discards the exact position of features, and upsampling alone cannot restore it. The skip connections pass the full-resolution encoder features directly to the decoder, so the network can combine *what* is present (from the coarse levels) with *where* it is (from the fine levels).

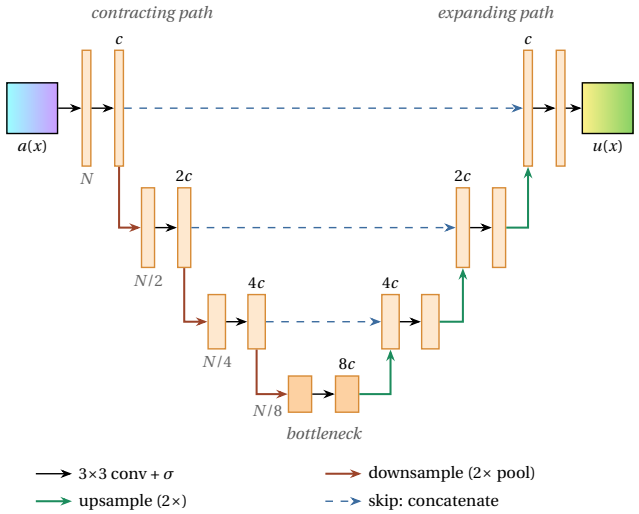


Figure 2. U-Net architecture with $S = 3$ downsampling steps. The contracting path (left) halves the resolution N and doubles the channel count c at each level; the expanding path (right) reverses this. Dashed skip connections concatenate encoder features with the upsampled decoder features of the same resolution, which restores the spatial detail lost by pooling.

Why it captures sharp structure. The U-Net provides an explicit hierarchy of local receptive fields. Its finest levels operate on the full grid with local kernels, so edges can be represented directly in physical space instead of through a truncated set of global modes. Its intermediate levels give it receptive fields at every scale between a few cells and the full box, which matches the broad range of bubble sizes. Finally, skip connections let high-frequency detail bypass the bottleneck entirely. U-Nets are therefore strong baselines for field-to-field emulation whenever sharp, localized features matter.

Limitations. The convolution kernels are defined in grid cells, so their physical support changes with resolution. A fully convolutional U-Net can often run on a different grid size, with padding where needed, but this does not make it a consistent discretization of the same physical operator. Pooling also introduces alignment dependence, so exact equivariance to arbitrary one-cell shifts is generally lost. Large-scale context passes through coarse features; whether this is sufficient depends on the task. The linear filters are shared spatially, while the nonlinear response remains input-dependent.

2.3 Hybrid architectures: U-FNO and related

The complementary strengths of the FNO (global, resolution-independent) and the U-Net (local, multi-scale) suggest combining them. The U-FNO of Wen et al. [12] does this inside the Fourier layer. In addition to the spectral path $\mathcal{F}^{-1}(R_\phi \cdot \mathcal{F}v)$ and the pointwise path Wv of Figure 1, each *U-Fourier layer* adds a small U-Net acting on v:

$$v_{\ell+1}(x) = \sigma\Big(W v_\ell(x) + \mathcal{F}^{-1}\big(R_\phi \cdot \mathcal{F} v_\ell\big)(x) + \mathrm{UNet}_\ell[v_\ell](x)\Big), \tag{8}$$

where UNet_ℓ denotes the U-Net path. The spectral path supplies global, low-k structure, while the U-Net path restores the high-k, localized detail that mode truncation removes. In the original application to multiphase flow, where saturation fronts are sharp, this substantially improved accuracy over the plain FNO [12].

The improvement comes at a cost that matters when comparing architectures. The U-Net path adds a large number of parameters, and a U-FNO is typically much larger than an FNO of the same width and depth. In our own comparison (Section 4), the U-FNO reference model has roughly two orders of magnitude more parameters than the matched models. Better accuracy can therefore not be attributed to the architecture alone. We treat that implementation as a dense reference baseline. Parameter matching alone is not enough to isolate architecture: inputs, losses, optimization and compute also need to be controlled. The convolutional path introduces cell-scale dependence into the U-FNO, even though its Fourier multipliers can be reused

across grids.

Other hybrids follow the same pattern of adding a local or multi-scale branch next to a global spectral one. They differ in which branch is used and where it is inserted, but they combine spatial branches with fixed learned filters and nonlinear maps, as in Equation (8).

2.4 Open problem: structure that adapts to the field

A useful question is whether an explicit, inexpensive modulation of the spatial kernel improves field prediction beyond what a stack of fixed filters and nonlinearities already learns. This is an architectural question, not a claim that standard nonlinear networks cannot adapt to their inputs.

Radiative transfer motivates such modulation: absorption depends on the intervening material, so a physical source–receiver response need not be stationary. The present targets, however, come from the semi-numerical prescriptions of `21cmFAST`, not a resolved transport calculation. Neither attention based only on endpoint features nor the rhizome's endpoint gates automatically encode optical depth along a ray. Intermediate information can enter through repeated contextual updates, but a physical transport interpretation requires separate evidence.

An operator suited to this problem should combine four properties: (i) global reach, so that interactions across the whole box are possible; (ii) state-dependent coupling, so that the effective connections change as the field evolves; (iii) a cost close to $\mathcal{O}(N \log N)$, so that three-dimensional grids remain tractable; and (iv) a parameterization that is independent of the grid, so that the discretization invariance of the FNO is kept. The rhizome operator, introduced in Section 3, is designed around these four requirements.

3 THE RHIZOME NEURAL OPERATOR

The rhizome neural operator combines a learned global convolution with source and receiver gates. Every update recomputes those gates from the hidden field and the fixed conditioning, then applies the same shared update rule. The name refers loosely to a rhizome, a spreading underground stem; the mathematical construction is a gated integral operator, not a model of plant growth or arbitrary graph rewiring.

This section derives the operator from the kernel integrals of Section 1, then describes the architecture (Figure 3), how it is evaluated efficiently, and the variants used in the experiments of Section 4.

3.1 Design principles

The design follows the four requirements formulated at the end of Section 2.4:

1. **Global reach.** Interactions between cells are carried by a learned kernel $\kappa_\theta(r - r')$ on the whole periodic domain, evaluated in Fourier space as in the FNO.

2. **State-dependent coupling.** Two learned *gates* modulate the kernel at the sending and the receiving end. They are recomputed from the current hidden state at every update, so the effective coupling between two cells depends on the field at both of them.

3. $\mathcal{O}(N \log N)$ **cost.** The source gate is absorbed into the transmitted value, and the receiver gate factors out of the integral. The interaction is then a pointwise multiplication, one convolution evaluated with the FFT, and another pointwise multiplication: the same asymptotic dependence on the number of cells as a Fourier layer, at fixed width and mode count.

4. **Grid-independent parameters.** The model has no convolution stencils, no pooling and no parameters tied to particular cells. The kernel is parameterized by its Fourier coefficients on the normalized domain, and everything else acts pointwise.

Two further choices complete the design. First, there is exactly *one* mechanism for spatial communication: the gated integral. There is no separate local or convolutional branch next to it, as in the U-FNO; lifting, the update proposal and decoding all act on single cells. Second, the same update block is applied T times with shared weights. The model is a recurrent dynamical system on the hidden state rather than a stack of distinct layers.

3.2 Mathematical background: from kernel integrals to the rhizome

The rhizome is best understood as a specific member of the family of kernel integral operators introduced in Section 1.2. This subsection derives it from the linear integral operator Equation (1) in three steps. First, the kernel is allowed to depend on the field it acts on. Second, that dependence is restricted to a separable form that keeps the FFT evaluation of the FNO. Third, the resulting operator is iterated. Throughout the continuum discussion, $D = \mathbb{T}^d$ is a periodic domain with normalized measure $\mathrm{d}\mu = \mathrm{d}r/|D|$, written as $\mathrm{d}r$ after rescaling to unit volume. We take the hidden field v and fixed conditioning $\boldsymbol{c}$ to be bounded measurable fields. The kernel is a finite matrix-valued Fourier series, so convolution and the pointwise maps are well defined on $L^\infty(D;\mathbb{R}^{d_v})$. Dependence on $\boldsymbol{c}$ is suppressed in some formulas for readability. These assumptions concern the ideal periodic operator; padding and LOS preprocessing require the qualifications given below.

A hierarchy of kernels. A broad class of kernel integral layers lets the kernel depend on position and the field at both endpoints:

$$(\mathcal{K}[v]\,v)(r) = \int_D \kappa\big(r, r', v(r), v(r'); \theta\big)\, v(r')\,\mathrm{d}r'. \qquad (9)$$

Different neural operators correspond to different restrictions of κ (Table 2). If κ depends on r and r' only, Equation (9) is the linear operator Equation (1), which includes Green's-function representations as a special case. If it depends only on $r - r'$, it is a convolution and becomes the Fourier layer Equation (3). If it depends on the field values, the operator becomes nonlinear in v, since v now enters both as the integrand and through the kernel. Self-attention [11] is the best-known example: there $\kappa \propto \exp\big(q(v(r)) \cdot k(v(r'))\big)$, normalized over r', and evaluating it on a grid costs $\mathcal{O}(N^2)$.

The rhizome sits between the Fourier layer and attention. It keeps a stationary kernel but multiplies it by two factors,

one that depends on the field at the receiving point and one that depends on the field at the sending point:

$$\kappa^{\mathrm{eff}}(r,r') = \mathrm{diag}\big(A(v(r))\big)\,\kappa_\theta(r-r')\,\mathrm{diag}\big(B(v(r'))\big), \quad (10)$$

with $A, B : \mathbb{R}^{d_v} \to (0,2)^{d_v}$ the receiver and source gates. Setting $A = B = 1$ recovers the ungated convolution on the transmitted value; the full update still includes its pointwise maps and damping. Compared with attention, the dependence on the two endpoints is *separable*: it is a product of a function of $v(r)$ and a function of $v(r')$. Linear attention also factorizes content dependence, but typically sums products of query and key features and includes normalization [4]. The rhizome instead scales each channel at the two endpoints around a signed, channel-mixing relative-position kernel. Its FFT efficiency comes from convolution structure, not from treating the spatial matrix as an arbitrary low-rank attention matrix.

Operator form. Write M_a for the multiplication operator $(M_a f)(r) = a(r)\,f(r)$, applied channel by channel, and C_κ for the convolution $(C_\kappa f)(r) = \int_D \kappa(r-r')f(r')\,\mathrm{d}r'$. The rhizome message then reads

$$\boldsymbol{m} = M_{A(v)}\, C_{\kappa_\theta}\, M_{B(v)}\, V(v), \quad (11)$$

where V is the pointwise value map. Each factor is simple in one basis: the multiplication operators are diagonal in position space, and the convolution is diagonal in Fourier space, where $\widehat{C_\kappa f}(k) = \hat\kappa(k)\hat f(k)$. The rhizome therefore alternates between the two bases: multiply in position space, transform, multiply in Fourier space, transform back, multiply again. The convolution is block-diagonal in Fourier space: each retained mode still mixes channels. The calculation uses one forward and one inverse transform. Alternating bases also occurs in split-step methods, but that computational resemblance does not identify this product with a physical propagator or a splitting of a known evolution equation.

For fixed gates, the factorization gives a useful operator-norm bound. Since the gates lie in $(0,2)$, both multiplication operators have norm at most 2 on $L^2(D)$. By Parseval's theorem, the norm of the convolution is $\sup_k \|\hat\kappa_\theta(k)\|_2$. Hence

$$\|M_A\, C_{\kappa_\theta}\, M_B\, f\|_{L^2} \;\le\; 4 \max_{k\in\Lambda} \|\hat\kappa_\theta(k)\|_2\, \|f\|_{L^2}, \quad (12)$$

independently of the field that produced the gates. Here $\|\cdot\|_2$ is the matrix spectral norm. The gates cannot amplify a fixed transmitted field by more than this factor. The bound does not constrain the learned multiplier norm, the size of $V(v,\boldsymbol{c})$, or derivatives through the gates. It is therefore not a contraction or gradient-stability guarantee for the nonlinear rollout.

Mode coupling: smooth transport, sharp modulation. As discussed in Section 1.4, the convolution output is restricted to Λ, although the full FNO output need not be. The rhizome adds explicit multiplication before and after that convolution. A product in position space is a convolution in Fourier space, so the Fourier transform of Equation (11) is

$$\begin{aligned} \hat{\boldsymbol{m}}(k) &= \sum_q \mathrm{diag}\big(\hat A(k-q)\big)\,\hat\kappa_\theta(q)\,\widehat{(B\odot V)}(q), \\ \hat\kappa_\theta(q) &= 0 \quad \text{for } q \notin \Lambda. \end{aligned} \quad (13)$$

The kernel transports only $q \in \Lambda$. Before convolution, source multiplication mixes frequencies according to $\widehat{B\odot V}(q) = \sum_p \hat B(q-p)\odot \hat V(p)$; some high-frequency content can thereby contribute to retained modes. Receiver multiplication can then generate output frequencies outside Λ. This is ordinary Fourier mode mixing in the continuum, distinct from sampling aliasing on a grid. It does not guarantee that all input frequencies are preserved or transferred. If the receiver gate is sharp, for example close to zero inside an ionized region and close to two outside, a smooth message is multiplied by a sharp mask, and the result has a sharp edge exactly where the gate changes. The rhizome can thus carry information smoothly over long distances with few modes while placing sharp structure where the state of the field puts it. In a trained model the receiver gates do vary strongly in space (Figure 4); whether this is what produces sharp edges in the output has not been tested directly, and Equation (13) shows only that the mechanism is available.

Iteration as a path series. A single application of Equation (10) couples r' to r through the gates at the two endpoints only. Iteration adds paths. Ignoring for a moment the nonlinearity and the damping of the update, and freezing the gates, the recurrence becomes $v^{t+1} = c + \mathscr{K} v^t$ with $\mathscr{K} = M_A C_{\kappa_\theta} M_B$. After T steps this gives

$$v^T = \sum_{n=0}^{T-1} \mathscr{K}^n c \;+\; \mathscr{K}^T v^0, \quad (14)$$

a finite expansion that holds for any bounded $\mathscr{K}$. It converges to $(I-\mathscr{K})^{-1}c$ as $T\to\infty$ under a sufficient condition such as $\|\mathscr{K}\| < 1$, which also makes the final term vanish. Invertibility of $I-\mathscr{K}$ alone is not enough to guarantee convergence of this series. With frozen gates this is a linear Fredholm equation; no resolvent formula is asserted for the actual state-dependent network. Each term of the series is a sum over paths: the second-order term, for example,

$$\begin{aligned} (\mathscr{K}^2 f)(r) = \int_D\!\int_D & \mathrm{diag}\,A(r)\,\kappa_\theta(r-r_1) \\ & \times \mathrm{diag}\big(B(r_1)\odot A(r_1)\big) \\ & \times \kappa_\theta(r_1-r_2)\,\mathrm{diag}\,B(r_2)\,f(r_2)\,\mathrm{d}r_2\,\mathrm{d}r_1, \end{aligned} \quad (15)$$

couples r_2 to r through every intermediate point r_1, weighted by the gates there. This shows how intermediate states can influence a multi-update computation. These are algebraic paths through the interaction graph, not necessarily spatial rays. Small gates suppress paths that visit a region, but the global kernel can connect the endpoints directly or through other regions. Iteration therefore does not establish physical shadowing, causal propagation or radiative transfer. In the nonlinear network, gates and local derivatives change at each update; the finite expansion is an explanatory simplification, not an exact solution formula for the trained model.

Continuous-time view. The update of Algorithm 1, $\boldsymbol{h}^{t+1} = (1-\Delta t)\boldsymbol{h}^t + \Delta t\,\boldsymbol{p}^t$, is the forward Euler discretization of the nonlocal evolution equation

$$\partial_\tau \boldsymbol{h}(r,\tau) = -\boldsymbol{h}(r,\tau) + \tanh\Big(\Phi\big(\boldsymbol{h}(r,\tau),\,(\mathscr{K}[\boldsymbol{h}]\,V(\boldsymbol{h}))(r),\,\boldsymbol{c}(r)\big)\Big), \quad (16)$$

Table 2. Kernel integral operators ordered by what the kernel may depend on. N: number of grid points.

Operator	Kernel κ	Linear	Stationary	Cost
general fixed kernel	$\kappa(r, r')$	yes	no	$\mathcal{O}(N^2)$
Fourier layer	$\kappa(r - r')$	yes	yes	$\mathcal{O}(N \log N)$
rhizome	$\operatorname{diag} A(v(r))\, \kappa_\theta(r - r')\, \operatorname{diag} B(v(r'))$	no	no	$\mathcal{O}(N \log N)$
attention	$\kappa(v(r), v(r'))$, normalized	no	no	$\mathcal{O}(N^2)$

integrated from $\tau = 0$ to $\tau = T\Delta t$ (3 in the experiments), with Φ the pointwise proposal network. Equations of this form, a decay term plus a saturating function of a spatially integrated input, are the neural field equations of computational neuroscience [1], in which $\partial_t u = -u + \int w(r - r')\, S(u(r'))\, \mathrm{d}r' + I(r)$ describes the activity of a continuous sheet of neurons. The rhizome generalizes that picture with many channels, a learned kernel and connections gated by the state. This gives an ODE interpretation on a function space, related to the neural-ODE viewpoint [3]. The variable τ is computational time, not cosmic time or redshift. Training one fixed-step rollout does not establish that changing Δt at fixed $T\Delta t$ preserves its predictions. The bound on the hidden state follows directly: if $|\boldsymbol{h}| \le 1$ componentwise, then so is $(1 - \Delta t)\boldsymbol{h} + \Delta t\, \boldsymbol{p}$ for $|\boldsymbol{p}| \le 1$ and $0 < \Delta t \le 1$.

Discretization. On a uniform grid of N points on the unit-volume domain, the integral in Equation (9) becomes the sum Equation (4) with weights $w_j = 1/N$. For a symmetric rectangular retained set Λ with $|k_\ell| < m_\ell$ in direction ℓ, the implementation requires $n_\ell \ge 2m_\ell$ grid points in each direction, keeping the modes distinct and below Nyquist. With the kernel represented by those modes, $\kappa_\theta(r) = \sum_{k \in \Lambda} \hat{\kappa}_\theta(k)\, e^{2\pi i k \cdot r}$, this sum is evaluated *exactly* by the FFT:

$$\frac{1}{N} \sum_{j=1}^{N} \kappa_\theta(r_i - r_j)\, g(r_j) = \sum_{k \in \Lambda} \hat{\kappa}_\theta(k)\, \hat{g}_k\, e^{2\pi i k \cdot r_i}, \qquad \hat{g}_k = \frac{1}{N} \sum_{j=1}^{N} g(r_j)\, e^{-2\pi i k \cdot r_j}, \tag{17}$$

provided the grid satisfies that mode-resolution condition. For this fixed finite kernel, the FFT introduces no approximation beyond floating-point arithmetic to the stated grid sum. Relative to a continuum target, there are also sampling and quadrature errors, the finite spectral representation, and aliasing in the surrounding nonlinear maps. This identity is what the numerical verification in Section 3.6 checks, by evaluating the left-hand side directly over all pairs.

3.3 State-dependent integral interaction

Each grid cell i at position r_i carries a hidden state $\boldsymbol{h}_i^t$ of d_v channels and a fixed conditioning vector $\boldsymbol{c}_i$, built from the input density and the global parameters. At update t, cell i receives the message

$$\boldsymbol{m}_i^t = \frac{1}{N} \sum_{j=1}^{N} \operatorname{diag}\big(A_\theta(\boldsymbol{h}_i^t, \boldsymbol{c}_i)\big)\, \kappa_\theta(r_i - r_j) \times \operatorname{diag}\big(B_\theta(\boldsymbol{h}_j^t, \boldsymbol{c}_j)\big)\, V_\theta(\boldsymbol{h}_j^t, \boldsymbol{c}_j), \tag{18}$$

where

- V_θ builds the *value* that cell j transmits from its state and conditioning,
- B_θ is the *source gate*: how strongly cell j transmits, per channel,
- $\kappa_\theta(r_i - r_j) \in \mathbb{R}^{d_v \times d_v}$ is a learned, periodic, channel-mixing kernel that carries the signal across the separation $r_i - r_j$,
- A_θ is the *receiver gate*: how strongly cell i accepts what arrives, per channel,
- N is the number of grid cells, so that $1/N$ is the uniform quadrature weight on the normalized periodic box.

Read from right to left, Equation (18) describes a transmission: cell j prepares a value, decides how loudly to send it, the kernel carries it over the distance $r_i - r_j$, and cell i decides how much of it to take in. It is the quadrature of the operator Equation (11), with the effective kernel Equation (10) evaluated at the current state $(\boldsymbol{h}^t, \boldsymbol{c})$. This is the key difference from the FNO: the base kernel κ_θ is stationary, but the effective kernel depends on the field at both ends of every connection and changes from update to update.

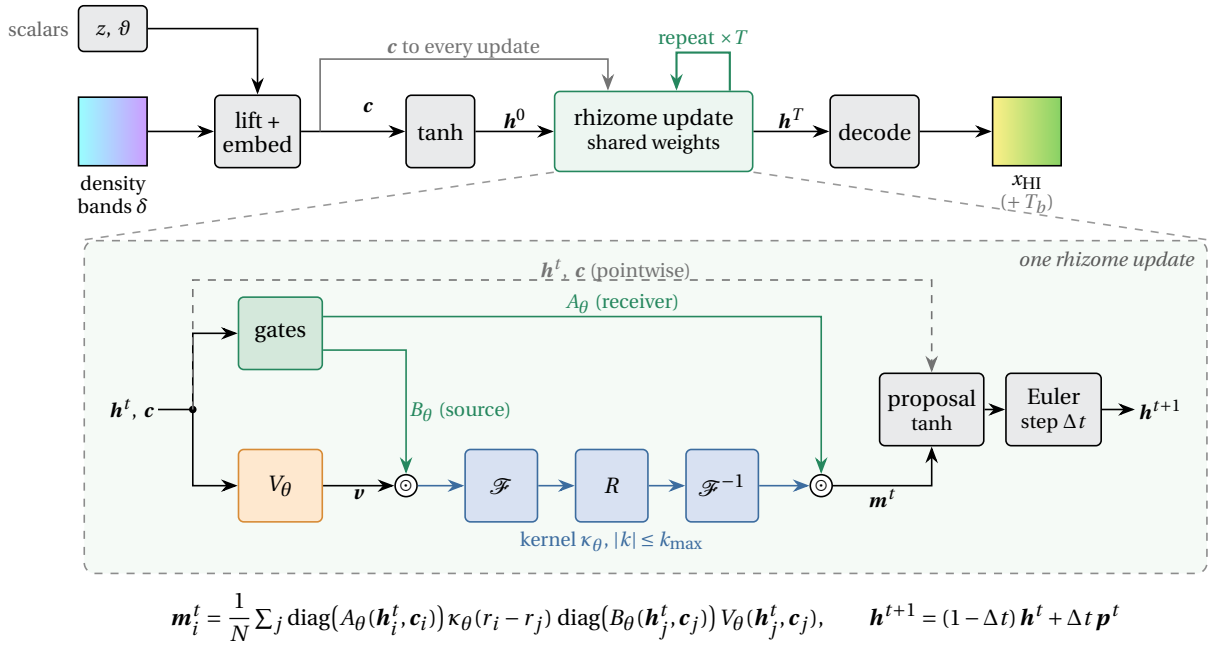
The gates are deliberately simple. The source gate acts on each transmitted value; the receiver gate factors out of the sum in Equation (18). This is what keeps the cost at that of a Fourier layer: the network never forms the $N \times N$ matrix of pairwise couplings, which for a 140×140 slice alone would have about 3.8×10^8 entries per channel pair. The price is expressivity. The coupling between two cells is a product of a sender factor and a receiver factor around a fixed kernel, not an arbitrary function of both states as in attention. In particular, a single update cannot make the coupling between r and r' depend on the cells *between* them. Repeated updates can incorporate intermediate contextual information, as illustrated by Equation (15), but whether they learn a particular transport effect must be tested.

3.4 Gating: source and receiver factors

A single pointwise network maps $[\boldsymbol{h}^t, \boldsymbol{c}]$ to $2d_v$ logits per cell, which are split into the source and receiver gates and passed through $2\,\mathrm{sigmoid}(\cdot)$. Each gate lies in $(0, 2)$ and equals one at zero logit. Zero-initializing the gate logits would recover an ungated interaction initially; the present implementation uses the default pointwise-layer initialization, so its initial gates need not all equal one.

Three properties of the gates are worth stating explicitly:

- **They are recomputed at every update** from the previous state and the fixed conditioning. A change in the hidden state can alter transmission and reception at the next update. The hidden channels are latent features, not an ionization fraction evolving in physical time.



$$m_i^t = \frac{1}{N}\sum_j \mathrm{diag}\big(A_\theta(h_i^t, c_i)\big)\,\kappa_\theta(r_i - r_j)\,\mathrm{diag}\big(B_\theta(h_j^t, c_j)\big)\,V_\theta(h_j^t, c_j), \qquad h^{t+1} = (1-\Delta t)\,h^t + \Delta t\,p^t$$

Figure 3. Rhizome Neural Operator architecture. *Top:* the input density bands and the global scalars (redshift z, simulation parameters ϑ) are lifted and embedded into a conditioning field $\boldsymbol{c}$, which initializes the hidden state and is fed into every update. One update block with shared weights is applied T times; a pointwise decoder maps the final state to x_{HI} (and, in the multi-field model, to T_b). *Bottom:* one update. The gates produce a source factor B_θ and a receiver factor A_θ from the current state. The source-gated value is convolved with the kernel κ_θ in Fourier space and multiplied by the receiver gate, giving the message $\boldsymbol{m}^t$. A pointwise proposal combines message, state and conditioning, and an explicit Euler step updates the state.

- **Source and receiver are separate outputs.** Interactions may be asymmetric: cell j can influence cell i strongly while i barely influences j. Since the kernel is not constrained to be positive, interactions may also be signed.
- **They are channel-wise.** Each of the d_v channels has its own gate, so different channels can carry different kinds of information with different reach. Channels are mixed only by the kernel κ_θ and by the pointwise networks.

No physical constraint is imposed: the operator does not enforce reciprocal couplings, conservation of photons or any form of radiative transfer. Whatever physical interpretation the gates acquire is learned from the data.

What trained gates do. Figure 4 inspects a rhizome trained on synthetic toy lightcones (32 × 32 slices, long single-seed run, four updates). It shows a validation slice at $z = 10.55$ with mean neutral fraction 0.50 and one receiver cell on an ionization front. The base kernel is a compact, translation-invariant blob (b). The receiver gate varies strongly across the slice (c). The source gate varies only between 1.04 and 1.12 in channel RMS (d), because individual channels move in opposite directions and partly cancel. Panels (e–h) show how the gates re-weight the connections into the receiver cell. They boost a ring of sources at intermediate distance and damp distant ones by up to about 20%, while the mean gain grows from 0.95 to 1.67 over the four updates. Once this gain is normalized out, the shape of the effective connection map changes by only about 4% between the first and the last update: this model re-weights its connections moderately rather than re-wiring them. These are gains on the transmitted value for one slice and one cell, not a measure of causal influence, which would need an input-Jacobian or perturbation study.

3.5 Recurrent time-stepping

The hidden state is initialized from the conditioning and updated T times with the same weights (Algorithm 1). Each update forms the message, combines it with the state and the conditioning in a pointwise *proposal*, and takes an explicit Euler step of size Δt towards that proposal. The final state is decoded pointwise into the x_{HI} logit. Applying a sigmoid restricts x_{HI} to $(0, 1)$ at finite logits; exact neutral and ionized endpoints are approached as limits.

Algorithm 1 Rhizome operator forward pass

Require: density bands δ, redshift z, parameters ϑ
$\boldsymbol{c} \leftarrow \mathrm{lift}(\delta) + \mathrm{embed}(z, \vartheta)$
$\boldsymbol{h}^0 \leftarrow \tanh(\boldsymbol{c})$
for $t = 0, \ldots, T-1$ **do** ▷ shared weights
  $(\boldsymbol{s}, \boldsymbol{r}) \leftarrow 2\,\mathrm{sigmoid}\big(\mathrm{gates}([\boldsymbol{h}^t, \boldsymbol{c}])\big)$ ▷ source, receiver
  $\boldsymbol{v} \leftarrow V_\theta([\boldsymbol{h}^t, \boldsymbol{c}])$
  $\boldsymbol{m}^t \leftarrow \boldsymbol{r} \odot \mathscr{F}^{-1}\big(R \cdot \mathscr{F}(\boldsymbol{s} \odot \boldsymbol{v})\big)$
  $\boldsymbol{p} \leftarrow \tanh\big(\mathrm{pointwise}([\boldsymbol{h}^t, \boldsymbol{m}^t, \boldsymbol{c}])\big)$
  $\boldsymbol{h}^{t+1} \leftarrow (1-\Delta t)\,\boldsymbol{h}^t + \Delta t\,\boldsymbol{p}$
end for
return $x_{\mathrm{HI}} = \mathrm{sigmoid}\big(\mathrm{decode}(\boldsymbol{h}^T)\big)$

Because the proposal is bounded by tanh and the update is a convex combination for $0 < \Delta t \le 1$, the hidden state stays bounded by one throughout the rollout. This prevents hidden-state amplitudes from diverging. It does not ensure small perturbations stay small, gradients remain well

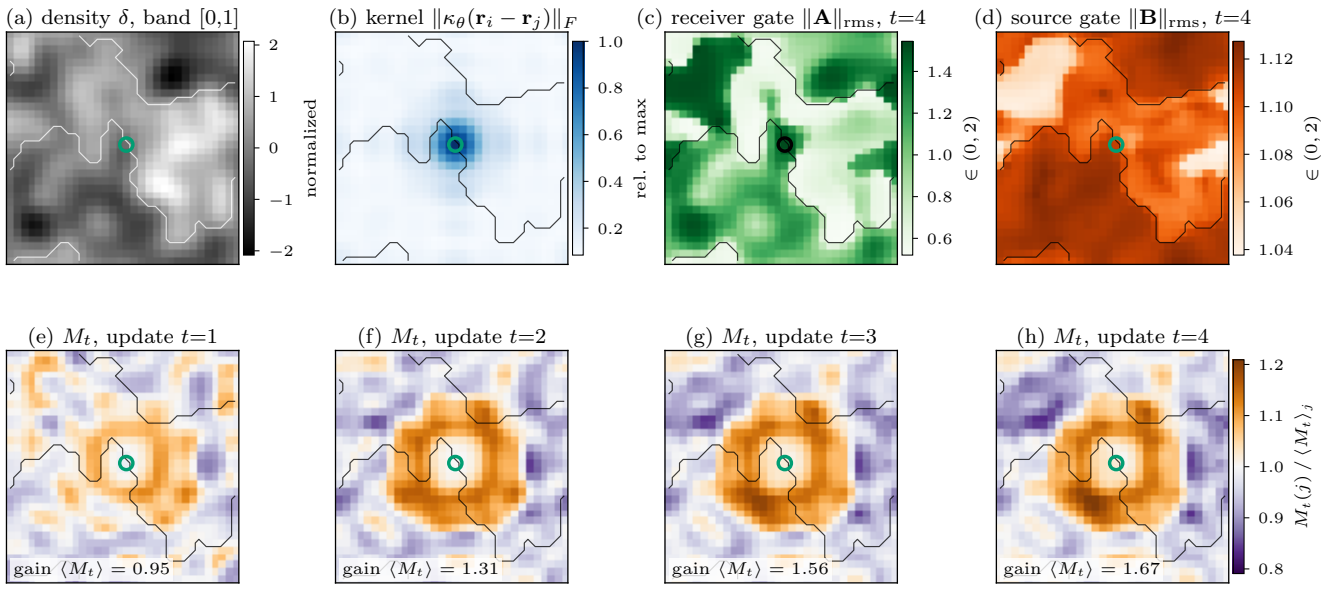


Figure 4. Learned interaction of a trained rhizome (synthetic toy lightcones, validation slice at $z = 10.55$, $\langle x_{\mathrm{HI}} \rangle = 0.50$, four updates). The ring marks the receiver cell i on an ionization front; black lines are the true $x_{\mathrm{HI}} = 0.5$ contour. (a) Input density band. (b) Norm of the learned kernel $\kappa_\theta(r_i - r_j)$, which is translation invariant. (c, d) Channel-RMS receiver and source gates at the last update. (e–h) Gate re-weighting of the connections into i, $M_t(j) = \|\mathrm{diag}(A_i)\,\kappa_\theta(r_i - r_j)\,\mathrm{diag}(B_j)\|_F / \|\kappa_\theta(r_i - r_j)\|_F$, divided by its spatial mean; the mean gain is printed in each panel. The gates boost a ring of sources at intermediate distance and damp distant ones by up to ≈20%, while the overall gain grows from 0.95 to 1.67 over the four updates. These are gains on the transmitted value, not output sensitivities.

conditioned, or the iteration approaches a fixed point. In practice the model is trained and evaluated at a fixed number of updates. All experiments in Section 4 use $T = 6$ and $\Delta t = 0.5$.

A sufficient condition for convergence. Let $\mathscr{P}_{\boldsymbol{c}}(h)$ denote the complete tanh proposal, including the gated message, at fixed conditioning. If it is Lipschitz with constant L_P on the invariant set $\|h\|_{L^\infty} \le 1$, the update $F_{\boldsymbol{c}}(h) = (1 - \Delta t)h + \Delta t\,\mathscr{P}_{\boldsymbol{c}}(h)$ obeys

$$\|F_{\boldsymbol{c}}(h) - F_{\boldsymbol{c}}(\tilde{h})\|_\infty \le \rho\,\|h - \tilde{h}\|_\infty, \qquad \rho = 1 - \Delta t + \Delta t L_P. \quad (19)$$

For $L_P < 1$ and $0 < \Delta t \le 1$, this is a contraction, giving a unique fixed point and geometric convergence. This is a sufficient condition, not a property enforced in training. In particular, reducing Δt does not make this bound contractive when $L_P > 1$.

For completeness, a bound on L_P can be written in terms of the implemented maps. Use the componentwise maximum norm and let

$$C_\infty = \max_a \sum_b \int_D |[\kappa_\theta(s)]_{ab}|\,\mathrm{d}s. \quad (20)$$

At fixed bounded $\boldsymbol{c}$, suppose A, B and V have Lipschitz constants L_A, L_B and L_V on the invariant state set, and $\|V(h, \boldsymbol{c})\|_\infty \le V_{\max}$. Adding and subtracting the three changed factors in $A \odot C_{\kappa_\theta}(B \odot V)$ gives

$$L_m \le 2C_\infty \big[(L_A + L_B)V_{\max} + 2L_V\big]. \quad (21)$$

For the affine pointwise proposal $\Phi = W_h h + W_m m + W_c \boldsymbol{c} + b$, the fact that tanh is 1-Lipschitz implies $L_P \le \|W_h\|_\infty + \|W_m\|_\infty L_m$. Finite Fourier kernels and the implemented pointwise maps satisfy these boundedness assumptions for bounded conditioning, so a finite rollout is a well-defined nonlinear map. Their learned norms need not satisfy $L_P < 1$. The same local Lipschitz structure supports the continuum ODE interpretation, while the invariant state bound prevents amplitude blow-up; neither statement establishes convergence to equilibrium.

Weight sharing makes the parameter count independent of T. It also makes the model behave like a learned dynamical system rather than a deep feed-forward network, which raises the question of what happens away from the training horizon. A small pilot on synthetic 32×32 lightcones answers this (Table 3). The tied rhizome improved modestly on an earlier additive recurrent model (Fourier message plus separate local message) at an equal parameter count, while an untied version with independent weights per update, at four times the parameters, was more accurate still. Running the tied model for more updates than it was trained with degraded its accuracy, although the hidden state stayed bounded. The number of updates is therefore part of the trained model, not a free accuracy knob.

3.6 Discretization and spectral evaluation

The FFT in Figure 3 is not a separate component that predicts anything; it is simply an efficient and exact way to evaluate the sum over j in Equation (18) (Equation (17)). The kernel κ_θ is parameterized by its Fourier coefficients $R(k)$ for $k \in \Lambda$, exactly as in the FNO (Equation (5)). Three details make this evaluation correct:

- **Quadrature.** With an orthonormal FFT pair, the forward and inverse transforms together already produce the $1/N$ weight of Equation (18). Adding a further factor for the cell area would make messages shrink as the resolution grows.

Table 3. Architecture pilot on synthetic 32 × 32 lightcones (single seed, CPU; 512 training slices, 64 test slices). *Top:* rhizome against the additive recurrent baseline. *Bottom:* the tied rhizome evaluated at different numbers of updates; it was trained with four. Mixed slices have mean x_{HI} between 0.05 and 0.95. This toy problem is not a physical simulation; the table characterizes the architecture, not its accuracy on 21 cm data.

Model	Params	RMSE, mixed
Rhizome, tied	15.6 k	0.196
Additive recurrent	15.9 k	0.207
Rhizome, untied	60.3 k	0.181

Updates T	RMSE, mixed
1	0.260
2	0.205
4 (trained)	**0.196**
8	0.265

- **Real kernel.** A real Fourier series requires $R(-k) = \overline{R(k)}$. The 2-D implementation explicitly enforces the self-conjugate-axis constraints. The 3-D real inverse FFT returns a real field, but projects unconstrained self-conjugate-plane coefficients onto their representable part; a real output alone does not show that all learned coefficients are independent or Hermitian-consistent.
- **Verification.** The periodic 2-D FFT implementation was checked against an independent calculation of the left-hand side of Equation (17) that sums the retained Fourier series over all source–receiver pairs explicitly, without any FFT. Outputs agree to about 10^{-15}, and so do the gradients with respect to inputs and parameters.

The architecture inherits the FNO's discretization properties (Section 1.4). Translation equivariance follows from convolution and shared pointwise maps, rather than from parameter count alone. For a periodic shift $S_a f(r) = f(r - a)$, convolution satisfies $C_{\kappa_\theta} S_a = S_a C_{\kappa_\theta}$, and the gate and value maps commute with jointly shifting h and $\boldsymbol{c}$. Consequently the periodic update and its pointwise decoder are equivariant when spatial conditioning is shifted too. Absolute coordinates, unshifted positional inputs, boundary padding and cropping can break this symmetry. It is exact for circular grid shifts in the periodic implementation; it is not automatically a statement about continuous sub-cell shifts. The same weights can be applied to any grid that represents the same physical box satisfying the per-direction mode-resolution condition of Equation (17). The caveats also carry over: the gates and the proposal are nonlinear and alias on coarse grids, so transfer across resolutions has to be validated rather than assumed. It has not been tested in this work. Moreover, the slice-model LOS bands use fixed cell offsets and cell-count averages: at a new resolution their physical reach changes. Preserving the complete predictor would require rebuilding those channels over the same physical intervals, not simply passing a finer array to the unchanged preprocessing.

3.7 Model variants

The same operator was used in four configurations, summarized in Table 4. They differ in what the model sees along the line of sight (LOS) and in which fields it predicts.

Padded-domain normalization in 3-D. The uniform weight $1/N$ in the continuum derivation counts cells on the periodic integration grid. In the 3-D implementation, zero-padding the transmitted field changes that count to $N_{\mathrm{pad}} = n_x n_y (Z + P)$, where Z is the physical window length and P the added FFT padding. Cropping the result does not change the denominator. For example, a unit zero-mode multiplier and a constant physical source field give $Z/(Z + P)$, rather than one. This is a valid normalized padded-domain convention. With fixed Z and P, the learned kernel can absorb its constant gain; it is not the same convention as normalizing by the physical-window volume. Converting to that latter convention would multiply the cropped message by $(Z + P)/Z$ and would change existing checkpoint predictions. Neither a normalization correction nor a halo removes the separate signed-offset collisions caused by insufficient padding. Resolution transfer must preserve the physical Fourier period as well as the integration measure.

2-D slices with LOS context. The first variant predicts one transverse 140 × 140 slice of a lightcone. Its input is the density on that slice and on the neighbouring slices: single slices within ±3 cells and band means further out, up to ±32 cells, for 13 density channels in total. The conditioning scalars are $\log(1 + z)$ and the eleven simulation parameters.

3-D native windows. The second variant works directly in three dimensions on windows of the native lightcone. In the implemented window convention, 256 cells is the total LOS input length, including a 32-cell halo at each end; the supervised core has 192 cells. The extra 64 zero-padded FFT cells are separate from those physical halo cells, giving a periodic FFT length of 320. It uses anisotropic mode counts (36 in each transverse direction, 24 along the LOS) and a rank-48 factorized spectral kernel to keep the parameter count at 5.8 M. A code review found that this padding is too short to separate all physical LOS offsets: displacements of +160 and −160 cells fall on the same residue of the padded circle, so the base kernel cannot treat them differently. The gates can still modulate these couplings, but the 3-D kernel is a constrained rather than a fully general LOS interaction.

Slice-wise reconstruction of 3-D cones. The third variant applies the 2-D slice model to every native slice of a lightcone, each with its own ±32-cell LOS context, and restacks the predictions into a full 3-D cone (Figure 5). In the language of windowed inference this is maximal overlap: window one slice, stride one slice. The model never sees the full LOS jointly; it only needs to know what lies within about 46 Mpc of each slice. For the tested suite, this context gives the best reported trade-off, and it avoids both the padding issue above and the memory cost of 3-D windows.

Multi-field prediction. The final variant predicts both x_{HI} and the 21 cm brightness temperature T_b. It receives 13 LOS bands each of density and LOS velocity. Its scalars are $1/(1 + z)$, the eleven parameters and the band means

Table 4. Rhizome variants used in the experiments. LOS: line of sight.

Variant	Input	Output	Kernel
2-D slice	one transverse slice + 13 LOS density bands (±32 cells)	x_{HI} on the slice	2-D, 24 modes
3-D window	256 LOS cells in total, including 2 × 32 halo	x_{HI} in the window	3-D, modes (36, 36, 24), rank-48 factorized
Slice-wise 2-D	2-D slice model applied to every slice of a cone	full 3-D x_{HI} cone, restacked	as 2-D slice
Multi-field slice-wise	density and LOS-velocity bands, global-history band means	x_{HI} and T_b (structured head)	as 2-D slice

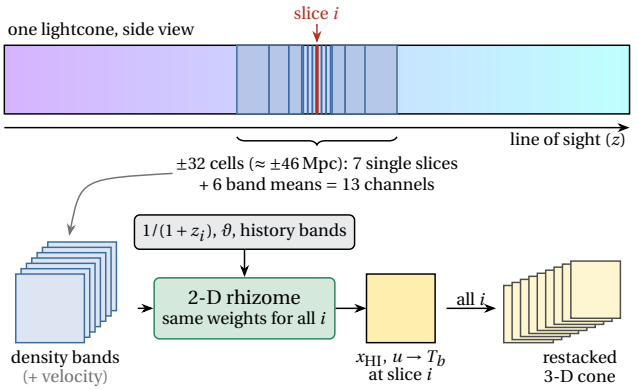


Figure 5. Slice-wise reconstruction of a 3-D lightcone. For each slice i along the line of sight, the 2-D rhizome receives 13 density channels: the seven single slices within ±3 cells and six band means out to ±32 cells (blue). In the multi-field model the same bands are built for the LOS velocity. Global scalars enter as conditioning. The prediction for slice i is computed independently, and the predictions for all slices are restacked into the 3-D cone.

of two emulated global histories, $\bar{x}_{\mathrm{HI}}(z)$ and $\bar{T}_b(z)$. The model outputs an x_{HI} logit and a spin variable u, and T_b is assembled by a *structured head* that encodes the known form of the signal:

$$T_b = T_0(z)\, x_{\mathrm{HI}}\, (1+\delta)\, F_v \left(1 - e^{u}\right), \tag{22}$$

where $T_0(z)$ is the redshift-dependent amplitude, F_v the correction for the LOS velocity gradient, and $1-e^u$ plays the role of the spin-temperature factor $1 - T_{\mathrm{CMB}}/T_S$. The prefactor $T_0(1+\delta)F_v$ is computed from the input fields using the same optically thin formula and clipped velocity-gradient convention as the baseline. It is not an exact radiative-transfer solution. The parameterization ensures a spin factor below one, compatible with positive T_S if $u = \log(T_{\mathrm{CMB}}/T_S)$; u is learned rather than supervised as a measured spin temperature. The implementation caps u from above for numerical control. The head links the two outputs through a known signal formula, but does not impose photon conservation or constrain arbitrary negative brightness-temperature excursions through the formula alone.

3.8 Assumptions and limitations

The rhizome replaces an explicit scale hierarchy with a particular factorization of the interaction. Its main assumptions are:

- **Periodicity.** The kernel lives on a periodic domain. This is exact in the transverse directions of the lightcones, which are periodic, but not along the LOS. The 2-D and slice-wise variants avoid the issue by treating the LOS through input bands; the 3-D variant relies on zero padding (Section 3.7).
- **Product-form coupling.** The effective kernel Equation (10) is a stationary kernel modulated at its two ends. It cannot represent couplings that depend jointly and arbitrarily on the states of both cells, or on the cells between them, within a single update.
- **No physics in the operator.** Reciprocity, conservation and radiative transfer are not imposed. Physics enters only through the inputs, the conditioning and, for T_b, the structured head.
- **Fixed rollout length.** The model is accurate at the number of updates it was trained with, not beyond (Table 3). There is no convergence or fixed-point guarantee.
- **Untested resolution transfer.** Although the parameters are grid-independent, all results in this report were obtained at a single resolution.

3.9 Relation to FNO, U-FNO and attention

Table 1 places the rhizome among the architectures of Section 2. Compared with the **FNO**, it keeps the global spectral kernel and its $\mathcal{O}(N \log N)$ cost, adds the gates that make the effective kernel state-dependent, and replaces the stack of independent layers by a weight-shared recurrence. The pointwise proposal takes over the role of the FNO's pointwise path W. Fixing the gates to one gives a recurrent spectral update. Recovering a particular standard FNO layer also requires compatible choices of the value map, proposal and residual step; removing the gates alone does not make the architectures identical.

Compared with the **U-Net and U-FNO**, it has no convolution stencils, no pooling and no explicit hierarchy of scales. Multi-scale behaviour has to come from the kernel, whose represented scales are set by Λ, and from the repeated updates. This keeps the parameterization independent of the grid, which the convolutional path of the U-FNO does not.

Compared with **attention**, it shares the central idea that the coupling between two points should depend on their content, but it restricts that dependence to a product of per-cell factors around a stationary kernel. This restriction is what brings the cost down from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$: the rhizome trades the full flexibility of pairwise attention for a form that can be evaluated with a single FFT pair per update.

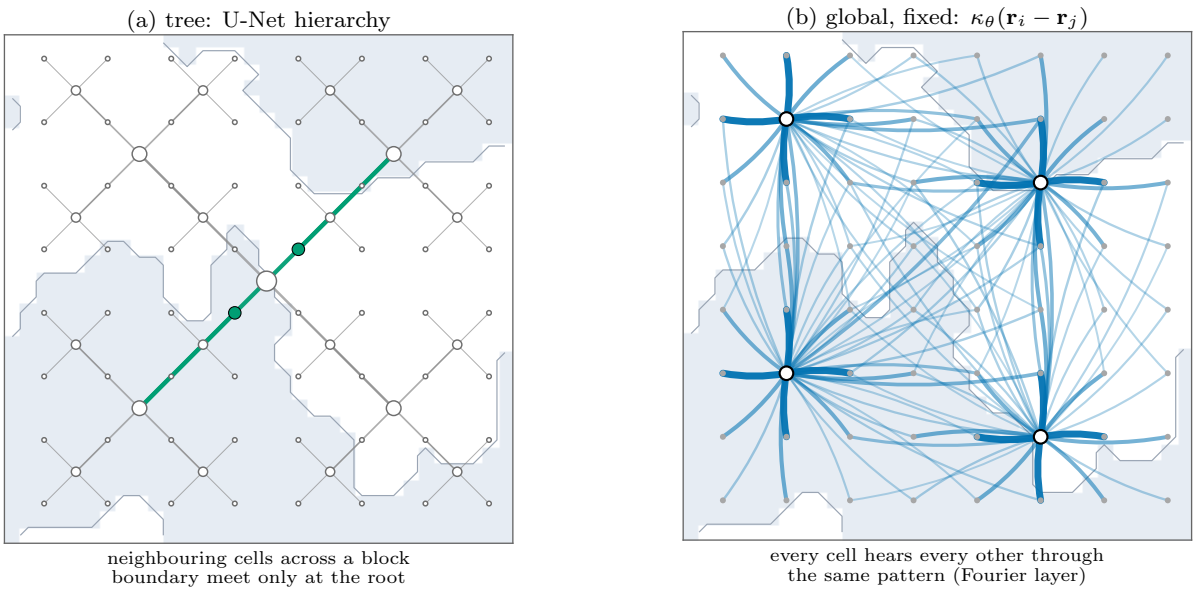


Figure 6. How cells exchange information, on an 8×8 lattice of cells of the toy slice of Figure 4 (shaded: true neutral region). (a) A U-Net-style hierarchy: information moves through coarse parent nodes, so two neighbouring cells on either side of a block boundary (green) meet only at the root. (b) An ungated global kernel, as in a Fourier layer, here the trained rhizome's own κ_θ with $A = B = 1$. Lines show the incoming connections of four cells; every cell is reached by every other, and the pattern is identical around each cell. (c) The trained rhizome at update 4, for the same four cells: edge $j \to i$ carries $\|\mathrm{diag}(A_i)\,\kappa_\theta(r_i - r_j)\,\mathrm{diag}(B_j)\|_F$, so it is scaled by both the receiver's gate (ring; channel RMS printed) and the sender's gate (dot shade; channel RMS, which spans only 1.04–1.12 and is stretched over the full shade range). The receiver gate scales a whole fan by 0.55–1.49; sender gates re-weight individual edges within a fan by 0.88–1.21. Width and opacity are relative to each panel's maximum; edges below 10% of it, and edges crossing the periodic boundary, are omitted.

Restrictions of the factorization. On a fixed grid and at a fixed state, flattening sites and channels gives $K_{\mathrm{eff}} = D_A K D_B$. The gate matrices are diagonal and strictly positive in exact arithmetic. They preserve the support, entrywise sign pattern and rank of K; they rescale existing couplings rather than create arbitrary edges. Very small sigmoid values can effectively suppress couplings numerically, and mixtures across channels can change the net message, but neither effect amounts to unconstrained graph learning. For a fixed channel pair, the modulation of a base-kernel entry is the separable product $A_a(r)B_b(r')$.

The frozen interaction matrix is also not the Jacobian of the nonlinear message. If $m(h) = M_{A(h)} C_{\kappa_\theta} M_{B(h)} V(h)$, its directional derivative at fixed conditioning is

$$\begin{aligned} Dm(h)[\eta] = {} & DA(h)[\eta] \odot C_{\kappa_\theta}(B \odot V) \\ & + A \odot C_{\kappa_\theta}\big(DB(h)[\eta] \odot V\big) \\ & + A \odot C_{\kappa_\theta}\big(B \odot DV(h)[\eta]\big). \end{aligned} \tag{23}$$

Here D denotes differentiation and undifferentiated factors are evaluated at $(h, \boldsymbol{c})$. Gate derivatives, proposal derivatives and later updates all contribute to output sensitivity. The connection plots therefore describe frozen-value transmission weights, not causal influence or an attribution of the accuracy gain.

Related gated and recurrent operators. The diagonal–convolution construction has close precedents outside PDE emulation. Hyena alternates input-controlled diagonal gates with long convolutions evaluated through the FFT [9]. AFT combines source weighting, positional interactions, normalization and receiver gating; its convolutional variant shares relative-position weights [15]. At the level of these operations, both overlap with the rhizome. The particular construction here uses a signed channel-mixing Fourier kernel and recomputes both gates from the evolving hidden field under a shared damped update, rather than holding input projections fixed within the block. The value of that feedback remains an empirical question.

Shared iterative updates also appear in implicit Fourier neural operators and nonlocal kernel networks [13, 14]. Their recurrence does not make their equilibrium or stability results applicable to this update. We claim neither the first gated convolution nor a general approximation theo-

rem for a fixed-width, fixed-mode, six-update rhizome. The general neural-operator approximation results of Kovachki et al. [5] do not by themselves prove such a result for this constrained architecture.

Figure 6 contrasts the three communication patterns on an 8×8 lattice of cells from the same toy slice as Figure 4. In a U-Net-style hierarchy (a), information moves through coarse parent nodes, so two neighbouring cells on opposite sides of a block boundary meet only at the root; skip connections and overlapping stencils soften this in a real U-Net. A global stationary kernel (b) connects every cell to every other cell with the same pattern around each of them. In the trained rhizome (c), each edge $j \to i$ is scaled by the receiver's gate and by the sender's gate. In this model the receiver dominates. Of the variance of the logarithm of the gate factor over all lattice pairs (0.125), 0.117 is explained by the receiver alone and 0.002 by the sender alone. A receiver's gate scales its whole incoming fan by 0.55–1.49, while sender gates re-weight individual edges within a fan by only 0.88–1.21. Total incoming strength therefore varies from cell to cell (coefficient of variation 0.32, against zero for the ungated kernel) and tracks the receiver gate almost exactly (correlation 0.997), while outgoing strength varies little (0.04). The connections are also directed: the median asymmetry $|w_{ij} - w_{ji}|/(w_{ij} + w_{ji})$ is 0.17. The weak sender effect is a property of this trained model, not of the architecture.

For a dense kernel with $M = |\Lambda|$ retained modes, one update costs $\mathcal{O}(d_v N \log N)$ for the FFTs, $\mathcal{O}(M d_v^2)$ for spectral channel mixing, and $\mathcal{O}(N d_v^2)$ for the pointwise maps at fixed relative conditioning width. A rollout multiplies these costs by T. Thus $\mathcal{O}(N \log N)$ describes spatial scaling at fixed width and mode count, not the complete dependence on model size. The dense spectral parameter count is $\mathcal{O}(M d_v^2)$.

The 3-D rank-r parameterization uses $R(k) = P_{\text{out}}\,\mathrm{diag}(w(k))P_{\text{in}}$, with projections shared across modes. Its spectral parameter count and mixing cost are $\mathcal{O}(Mr + d_v r)$ and $\mathcal{O}(M d_v r)$, respectively; the pointwise and FFT costs remain. This is a restriction on channel mixing, distinct from the separability of the endpoint gates. Weight sharing keeps update parameters independent of T, but computation still grows with the rollout length and stored activations can require $\mathcal{O}(TNd_v)$ memory without recomputation.

4 TRAINING AND RESULTS

This section reports how the rhizome was trained and how it compares with the best models of the existing `fno-21cm` pipeline on the task of predicting the neutral-fraction field x_{HI}, and later also the brightness temperature T_b, from the matter density of `21cmFAST` lightcones. The experiments followed the order of Table 4: 2-D slices, native 3-D windows, multi-field 3-D, and finally slice-wise reconstruction of full cones. Figure 7 summarizes the outcome.

4.1 Datasets

All data come from one suite of 6,600 `21cmFAST` [8] simulations (version 4.1.1), each with eleven sampled simulation parameters. Every simulation provides a 3-D lightcone of 140×140 transverse cells over a 200 Mpc box (1.43 Mpc per cell). The native line-of-sight length depends on cosmology and is between 2,074 and 2,919 cells at the same spacing. The lightcones contain the density, the neutral fraction, the brightness temperature and the LOS velocity, among other fields. Two properties of the data matter for the models:

- Each lightcone reuses the periodic 200 Mpc spatial realization along the LOS, with a tiling period of 140 cells. The fields can evolve with redshift along that direction, so this is repeated spatial sampling rather than an identical copy of the evolving field.
- Ionized cells have x_{HI} exactly zero; partially ionized values appear mostly where the lightcone interpolates between two simulation snapshots.

33 simulations (0.5%) contain non-finite T_b values and are excluded from every multi-field split. Table 5 lists the splits used.

The x_{HI} and multi-field splits are those of the `fno-21cm` pipeline. All comparisons with `fno` models below use identical test cones and the same evaluation code.

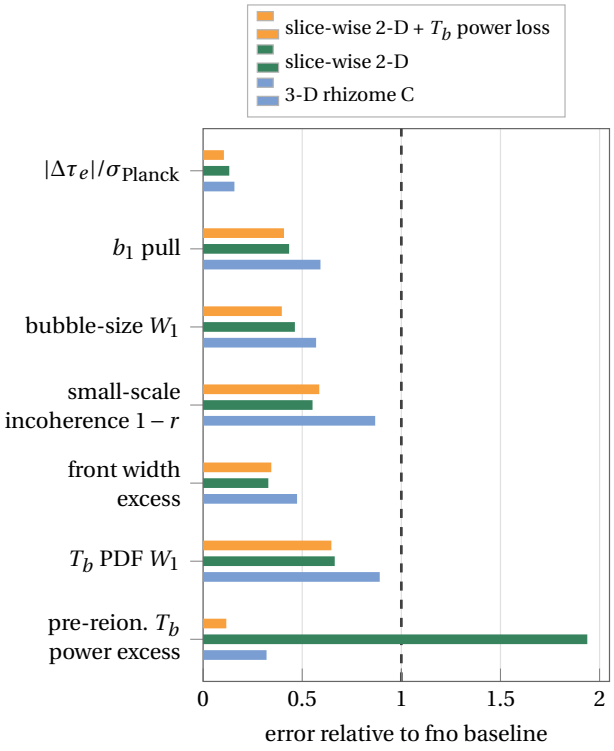


Figure 7. Headline results on 199 held-out multi-field test cones. Each bar is the error of a rhizome model divided by the error of the best `fno-21cm` multi-field model (`tbs+histtb`); values below the dashed line are better than the baseline. Front width is measured as the excess over the true front width of 2.77 Mpc. The reported slice-wise errors are lower on the listed metrics except the small-scale T_b power before reionization, which a power term in the loss reduces (Section 4.10). Numbers in Table 12.

4.2 Training protocol

Losses. x_{HI} is trained with binary cross-entropy (BCE) on the logit. In the multi-field models, T_b is added as a mean-squared error on the normalized field, through the structured head of Equation (22). The choice of BCE turned out to matter beyond the operator (Section 4.7). The `fno` base-

Table 5. Training, validation and test splits. All test sets are held out by simulation (cone), never by slice.

Split	Used for	Train / val / test cones	Samples
2-D slice cache	2-D sweep and first 2-D model	5,280 train (own split)	16 window + 2 background slices per cone
$x_{\rm HI}$ (fno split)	3-D rhizome; fair slice-wise model	1,600 / 200 / 200	3-D windows; or 48 + 6 slices per cone
Multi-field	3-D multi-field; slice-wise multi-field	1,594 / 199 / 199	as above, cleaned T_b
Multi-field, all cones	best slice-wise multi-field model	6,169 / 199 / 199	16 + 2 slices per cone (98.5 k)

lines train $x_{\rm HI}$ with a mean-squared error through a sigmoid, whose gradient vanishes when the prediction saturates.

Optimization and stopping. The learning rate is 2×10^{-3} throughout; the 3-D models use warmup and cosine decay. The 2-D models are stopped by a predefined plateau rule on the validation loss, and the checkpoint with the best validation score is kept. The 3-D models are trained for 20 epochs of 6,400 windows and then continued with a short *warm restart* (10 epochs at 5×10^{-4} with cosine decay), which improved test RMSE by 3–6%, against about 1% for a continuation at constant small learning rate. The 2-D models use the dihedral symmetries of the square slice as data augmentation.

Meaning of the neutral-fraction loss. BCE accepts targets in $[0,1]$, including partially ionized simulation cells; it need not mean that the output is a Bernoulli sample. For deterministic targets it acts as a bounded-fraction regression loss. If the compressed inputs leave some morphology unresolved, both BCE and squared error are minimized pointwise by the conditional mean of the target. BCE alone therefore does not eliminate averaging of uncertain boundaries or produce a distribution of possible fields.

Hyperparameters. Table 6 lists the main settings. Every model is a single training run with one seed.

4.3 Evaluation metrics

Per-voxel RMSE is reported, but it is a weak guide to physical quality: a model can lower its RMSE by blurring fronts. The evaluation therefore uses the physical suite of the `fno-21cm` pipeline, computed on full test cones. Most metrics are evaluated in the *active* stage of reionization, slices with mean neutral fraction between 0.05 and 0.95, and some also *before reionization*, $\bar{x}_{\rm HI}\geq0.98$, where T_b is in absorption. Table 7 lists them.

The cross-correlation coefficient between prediction and truth,

$$r(k)=\frac{P_{\rm pred\times true}(k)}{\sqrt{P_{\rm pred}(k)\,P_{\rm true}(k)}}, \tag{24}$$

deserves special attention. The power ratio measures whether a model produces the right *amount* of structure at each scale; $r(k)$ measures whether that structure is in the right *place*. In the earlier 3-D operator study of the `fno-21cm` pipeline, small-scale coherence predicted the bias in the bubble-size distribution across architectures (Spearman $\rho=0.88$), while small-scale power amplitude did not. Getting fronts in the right place matters more for bubble statistics than getting the right amount of small-scale detail.

4.4 Baselines

All baselines come from the `fno-21cm` pipeline and were trained there on the same splits:

- `cnn_whno_es` (3.5 M parameters): the `fno` 3-D $x_{\rm HI}$ model trained on native LOS windows. It has the U-FNO layout, a convolutional local path next to a learned global path, at a size comparable to the rhizome. Trained for 30 epochs (44.7 h on an A100).
- `es+histtb` and `tbs+histtb`: the corresponding multi-field models, which receive density, velocity and both emulated global histories. `tbs+histtb` additionally uses the structured T_b head and is the strongest `fno` model. It corresponds to rhizome variant C below and is labelled "UFNO" in the head-to-head figures.
- **2-D `fno` models** (LWF, WHNO, LocalFNO, U-FNO), which predict one slice from that slice alone.

The 203 M-parameter implementation of U-FNO [12] used in the earlier operator study (Section 2.3) is not used as a baseline here, because its capacity is not comparable.

4.5 2-D slices: line-of-sight context dominates

A sweep around a centre configuration (width 64, 24 modes, six updates) showed that the largest single factor is not a hyperparameter of the operator but the LOS context in the input (Figure 8). With only the target slice, the rhizome reaches a mixed-slice RMSE of 0.174. Adding the six neighbouring slices brings it to 0.138, and adding the band means out to ±32 cells to 0.112, a 36% reduction in total. By comparison, the other settings matter little: width 128 is 3% better than width 64 at four times the parameters, 12 or 48 modes are worse than 24, and three or ten updates are slightly worse than six. At equal compute, more training slices beat more width.

Table 8 compares the 2-D rhizome with the 2-D `fno` models on the same 7,036 validation slices. The rhizome reduces the error of the next best model by about 39%, and its ionization fronts are five times sharper. Most of this gap, however, is LOS information: the `fno` models see a single slice, and the single-slice rhizome (0.174) performs at their level. The 2-D comparison therefore shows what the input buys rather than what the operator buys. The fully trained 2-D model (width 128, plateau-stopped at 68 k steps) reached a test mixed-slice RMSE of 0.099 and an ionized-region IoU of 0.914.

Table 6. Main hyperparameters.

	2-D slice / slice-wise models	3-D window model
width d_v	64 (128 in early runs)	48
Fourier modes $k_{\max}$	24	(36, 36, 24), rank-48 factorized
updates T / step Δt	6 / 0.5	6 / 0.5
batch / sample	32 slices	native window, 140 × 140 × 256, including 2 × 32 halo
learning rate	2×10^{-3}, plateau-stopped	2×10^{-3} (warmup + cosine), 20 epochs + warm restart
precision	float32	bfloat16 autocast, float32 FFT
parameters	grow as d_v^2: 1.1 M (w32) to 18.3 M (w128)	5.8 M
training time	2.8–3.5 h (A100)	~12 h + 6 h restart (H200)

Table 7. Evaluation metrics. All are computed per test cone and summarized by the median over cones.

Metric	What it measures
$\lvert\Delta\tau_e\rvert/\sigma_{\text{Planck}}$	error in the CMB optical depth implied by the predicted global history, in units of the Planck uncertainty
$\lvert\Delta z\rvert$ at $\bar{x}$ = 0.25/0.5/0.75; duration	timing of reionization: error in the redshift at which the global neutral fraction crosses each value, and in the duration
x_{HI} PDF W_1	Wasserstein-1 distance between predicted and true one-point distributions of x_{HI}
partially ionized fraction	fraction of cells with intermediate x_{HI} (truth ≈ 0.09); measures hedging
b_1 pull; P_ε ratio	EFT diagnostics: error in the fitted linear bias in units of its uncertainty; ratio of stochastic power to the truth (ideal 1)
bubble-size W_1	Wasserstein-1 distance (Mpc) between mean-free-path bubble-size distributions
power error (large / intermediate / small scales)	$\lvert P_{\text{pred}}/P_{\text{true}} - 1\rvert$ of the transverse x_{HI} power spectrum in three k ranges; small scales mean $k > 1\,\text{Mpc}^{-1}$
$1 - r$, high k	small-scale incoherence: one minus the cross-correlation Equation (24), averaged over $k > 1\,\text{Mpc}^{-1}$
edge L^2 / H^1, ±2 Mpc	error (and error in gradients) in a ±2 Mpc band around true ionization fronts
front width	width of the predicted ionization fronts (Mpc), compared with the true width
T_b PDF W_1; T_b power deviation	distance between T_b distributions (mK); power difference in units of the single-slice sample variance (σ)

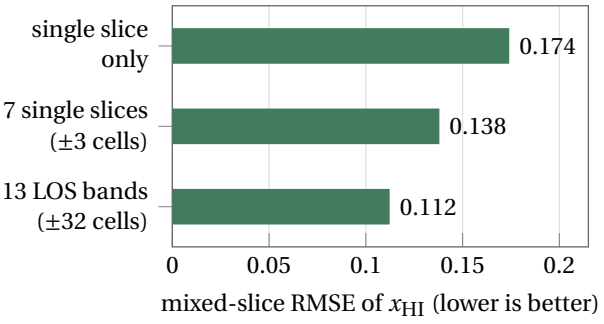


Figure 8. Effect of line-of-sight context on the 2-D rhizome (sweep, 15 k steps, one seed). Mixed slices have mean x_{HI} between 0.05 and 0.95.

Table 8. 2-D rhizome against 2-D `fno` models on 7,036 validation slices (rhizome: sweep run at 15 k steps). Only the rhizome sees LOS context. RMSE: slice RMSE; IoU: ionized-region intersection over union; front: front width (truth 3.0 Mpc).

Model	RMSE	IoU	Front (Mpc)
Rhizome (LOS bands)	**0.109**	**0.873**	**6.1**
`fno` LWF	0.178	0.755	32.6
`fno` WHNO	0.181	0.755	31.0
`fno` LocalFNO	0.184	0.742	–
`fno` U-FNO	0.195	0.722	–

4.6 3-D windows: x_{HI}

The 3-D rhizome sees the same information as the 3-D `fno` model: native LOS windows of the density field. This removes the LOS-input difference, although capacity, loss and optimization still differ, so it does not isolate the interaction law alone. On the 200 test cones (Table 9), the rhizome at width 48 reaches an RMSE of 0.0594 after the warm restart, against 0.0684 for `cnn_whno_es`, with 5.8 M against 3.5 M parameters. Narrower was better (width 64 gave 0.0628), and a learning rate of 3×10^{-4} was 13% worse than 2×10^{-3}.

The margin is much larger on the physical metrics than on RMSE. The rhizome is three times more accurate in τ_e and in the reionization midpoint, halves the bias error, and comes 40% closer in bubble sizes. Its small-scale power and coherence also hold up better at every stage of reionization.

Table 9. 3-D $x_{\rm HI}$ on 200 test cones. *Left:* test RMSE. *Right:* physical suite (medians over cones), rhizome w48 after the warm restart.

Model	Params	RMSE
rhizome w48, 20 ep.	5.8 M	0.0619
+ warm restart	5.8 M	**0.0594**
rhizome w64, + restart	–	0.0628
`cnn_whno_es`, 30 ep.	3.5 M	0.0684

Metric	Rhizome	`fno`
$\lvert\Delta\tau_e\rvert/\sigma_{\rm Planck}$	**0.019**	0.058
$\lvert\Delta z(\bar{x}=0.5)\rvert$	**0.028**	0.098
b_1 pull	**1.58**	3.41
P_ε ratio (ideal 1)	**0.89**	0.77
bubble-size W_1 (Mpc)	**0.81**	1.39
small-scale power error	**0.222**	0.336
edge H^1, ±2 Mpc	**0.137**	0.145
front width (Mpc)	**5.9**	7.5

4.7 3-D multi-field: inputs, heads and losses

The multi-field models predict $x_{\rm HI}$ and T_b from density and LOS velocity. Four 3-D rhizome configurations were trained to separate the contributions of inputs and output head (Table 10).

Table 10. 3-D multi-field ablation, test RMSE on 199 cones ($x_{\rm HI}$ / T_b in mK). History channels are the band means of the emulated global $x_{\rm HI}(z)$ and $T_b(z)$.

Run	Inputs / head	20 epochs	+ restart
A0	density → $x_{\rm HI}$ only	0.0602	0.0567
A	+ velocity, + T_b (plain head)	0.0648 / 9.49	0.0622 / 8.71
B	A + history channels	0.0509 / 6.26	0.0494 / 6.07
C	B + structured T_b head	0.0507 / 5.97	**0.0495 / 5.79**
`fno es+histtb`	= B	0.0577 / 6.63	
`fno tbs+histtb`	= C	0.0560 / 6.54	

- **History channels give the largest improvement in this ablation.** Going from A to B reduces the $x_{\rm HI}$ error by 21% and the T_b error by 34%.
- **Velocity and a T_b target alone do not help** $x_{\rm HI}$; A is even slightly worse than A0 (one seed each).
- **The structured head mainly helps** T_b: without it, T_b power is about 10% low at every stage. It also improves the reionization timing by 15–30%.
- **The density-only rhizome A0 already beats** `fno es+histtb` (0.0567 vs 0.0577), although the latter also receives velocity and both histories.

On the physical suite, model C beats `tbs+histtb` by a factor of 5 in τ_e (0.014 vs 0.076 σ), and is closer in b_1 (pull 1.52 vs 2.28), P_ε (0.935 vs 0.793), bubble sizes (W_1 0.71 vs 1.21 Mpc) and front width (4.53 vs 6.17 Mpc; truth 2.77). Before reionization, `fno` adds spurious small-scale T_b structure (+12.8σ against +3.9σ).

Influence of the loss on the performance gap. Both model families receive the same emulated global history, yet the global $\bar{x}_{\rm HI}(z)$ of the `fno` models sits 0.002–0.003 too low from $z \approx 8$ to 25. In truly neutral cells at $z > 15$ the mean error is −0.0020 and −0.0030 for the two `fno` runs against −0.00006 for the rhizome. A plausible contributor is the loss. For a neutral-fraction logit ℓ, prediction $p = \mathrm{sigmoid}(\ell)$ and target y, $\partial_\ell \mathcal{L}_{\rm BCE} = p - y$, whereas a squared-error loss gives $2(p-y)p(1-p)$. Both approach zero at a correct saturated prediction, but squared error has an additional sigmoid factor that slows correction near the endpoints. This is consistent with the observed neutral-cell offset; a loss-matched retraining is needed to establish its contribution. This uniform offset integrates into a τ_e bias. A fair comparison of τ_e and the global history therefore needs the `fno` models retrained with BCE. The offset alone does not establish an explanation for the morphology differences; the effect of changing the loss on those metrics also needs to be measured.

4.8 Slice-wise reconstruction and native 3-D models

The slice-wise variant (Figure 5) applies the 2-D model to every slice of a cone. For a fair comparison it was retrained on a new slice cache built from the same 1,600 training cones as the 3-D models, with otherwise unchanged settings. Table 11 compares it with the 3-D rhizome and with `fno` on all 200 test cones.

The slice-wise model wins on nearly every physical metric, trains about seven times faster than the 3-D rhizome, and has no jitter between slices: the roughness from slice to slice matches the truth to 2–3%, while both 3-D models are slightly over-smoothed along the LOS. Its advantage comes from front *placement*: small-scale incoherence drops from 0.20 to 0.12, the mid-scale power error by 60%, and the edge error by 21% (significant in a paired bootstrap over 175 cones). Width barely matters, so width 64 (2.8 h of training against 5.0 h) is the choice. Per cone, the 2-D model wins 11 of 13 inspected cones. It loses on the earliest reionizer, cone 27 ($z_{\rm mid} \approx 15.9$), where the longer context of the 3-D model helps slightly.

The error maps in Figure 9 show the qualitative difference. The slice-wise errors are thin rims along bubble walls, whereas the 3-D models, `fno` strongly and the 3-D rhizome weakly, miss whole ionized regions, and `fno` leaves a faint ionized haze over neutral gas.

For this simulation suite, a 2-D model with ±32 cells (≈ ±46 Mpc) of LOS context is more accurate and cheaper than the tested 3-D windows. This does not establish a universal physical interaction length. The periodic tiling, compressed context channels, padding restrictions and optimization budget may all contribute to the result.

4.9 Multi-field slice-wise results

The final model combines the slice-wise approach with the multi-field inputs and the structured head of model C: 13 LOS bands each of density and velocity, global-history band means as scalars, and $x_{\rm HI}$ and T_b as outputs (Section 3.7). It was trained on all 6,169 usable training cones (3.5 h on an A100) and evaluated on the 199 multi-field test cones.

Table 11. Slice-wise 2-D reconstruction against the 3-D models, $x_{\rm HI}$, 200 test cones, all trained on the same 1,600 cones.

Metric	2-D w128	**2-D w64**	3-D rhizome w48	`cnn_whno_es`
$\lvert\Delta\tau_e\rvert/\sigma_{\rm Planck}$	**0.011**	**0.011**	0.019	0.058
$\lvert\Delta z(\bar{x}=0.5)\rvert$	**0.011**	0.014	0.028	0.098
$x_{\rm HI}$ PDF W_1, active	**0.011**	0.013	0.021	0.030
b_1 pull	1.01	**0.85**	1.58	3.41
P_ε ratio (ideal 1)	**0.95**	0.94	0.89	0.77
bubble-size W_1 (Mpc)	0.59	**0.56**	0.81	1.39
power error, mid k	**0.062**	0.073	0.174	0.172
$1-r$, high k	**0.119**	0.120	0.199	0.181
edge H^1, ±2 Mpc	**0.109**	0.110	0.138	0.145
front width (Mpc; truth 2.65)	**4.03**	4.10	5.91	7.54

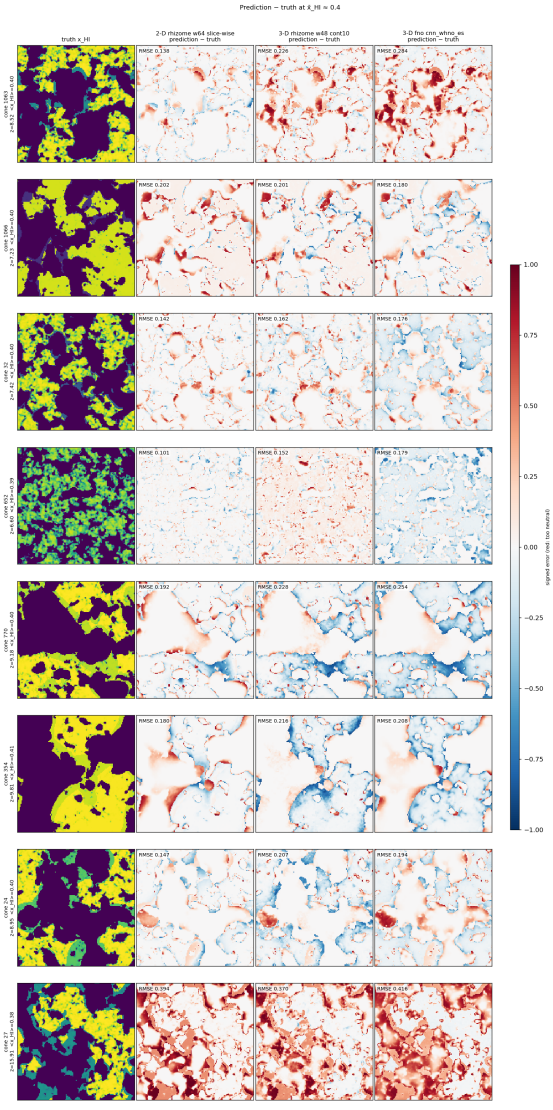


Figure 9. Signed $x_{\rm HI}$ error at $\bar{x}_{\rm HI}\approx0.4$ on eight test cones (red: predicted too neutral, blue: too ionized). Columns: truth, slice-wise 2-D rhizome (w64), 3-D rhizome (w48, warm restart), `fno` `cnn_whno_es`; per-panel RMSE in the corner. The bottom row is cone 27, the earliest reionizer, where all models predict ionized gas as too neutral.

Table 12 compares it with 3-D rhizome C and with the best `fno` model.

Against `fno`, the slice-wise model reduces the $x_{\rm HI}$ value-distribution error by 65%, the b_1 pull by 57%, the bubble-size error by 54% and small-scale incoherence by 45%, and its fronts are 3.9 Mpc wide against 6.2 Mpc. During reionization, its T_b distribution is 34% closer to the truth. On the eight test cones inspected individually, it beats `fno` on every cone, by 12–28% in full-cone RMSE. It also beats the 3-D rhizome C on every $x_{\rm HI}$ morphology metric (value distribution −39%, b_1 pull −27%, bubble sizes −19%, incoherence −36%, edges −19%), while τ_e and timing tie: with history channels, both rhizome models are already five to seven times better than `fno` there. Training on all cones instead of the 1,594-cone split improved per-cone RMSE by 3.6% ($x_{\rm HI}$) and 5.5% (T_b) in the median, on 86% and 75% of cones respectively.

Figure 11 shows the scale dependence. Both models lose small-scale power late in reionization, `fno` more so. The rhizome keeps $r>0.9$ to higher k at every stage, so the structure it does produce sits in the right place. Figure 12 puts this in the context of all models from the earlier 3-D operator study. In the plane of small-scale coherence against small-scale power error, the multi-field slice-wise rhizome sits apart from every `fno` model: $r(k>1)=0.90$ against 0.82–0.83, power error 0.17 against 0.31–0.34.

Figure 14 shows the signed error at $\bar{x}_{\rm HI}\approx0.4$ on eight test cones. As for the $x_{\rm HI}$-only model, the rhizome's errors are thin rims on bubble walls, while `fno` misses whole regions and its rims are wider.

Shared failure. Some test cones contain large, flat plateaus of intermediate neutral fraction ($x_{\rm HI}\approx0.3$–0.6) in the truth. No model reproduces them, and all overpredict the partially ionized fraction (0.13–0.18 against 0.09).

4.10 Brightness temperature before reionization

The slice-wise model has one clear weakness. Before reionization, when T_b is in absorption, it produces too much small-scale T_b power: $+24.8\sigma$ at $k=0.46\,{\rm Mpc}^{-1}$, against $+4.1\sigma$ for the 3-D rhizome and $+12.8\sigma$ for `fno` (Table 12). The excess is concentrated in quiet slices whose true small-scale power is tiny; the power summed over the cone is in fact slightly low (ratio 0.97). The working explanation is that the spin variable u is predicted slice by slice, and the per-pixel loss has no explicit term matching its fluctuation spectrum within a slice, whereas the 3-D model predicts it jointly along the LOS.

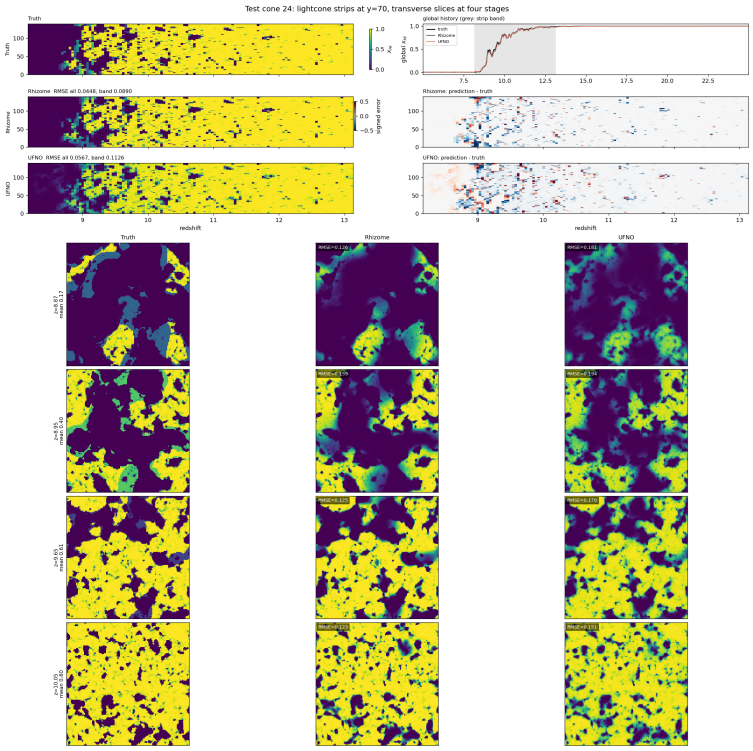


Figure 10. Test cone 24. *Top left:* x_{HI} lightcone strips for truth, rhizome (slice-wise multi-field) and the `fno` baseline (labelled UFNO). *Top right:* global history and signed error strips. *Bottom:* transverse slices at mean neutral fraction 0.17, 0.40, 0.61 and 0.80, with per-slice RMSE. Full-cone RMSE 0.045 (rhizome) against 0.057 (`fno`). The `fno` fronts are blurred, and it leaves a faint haze inside ionized regions.

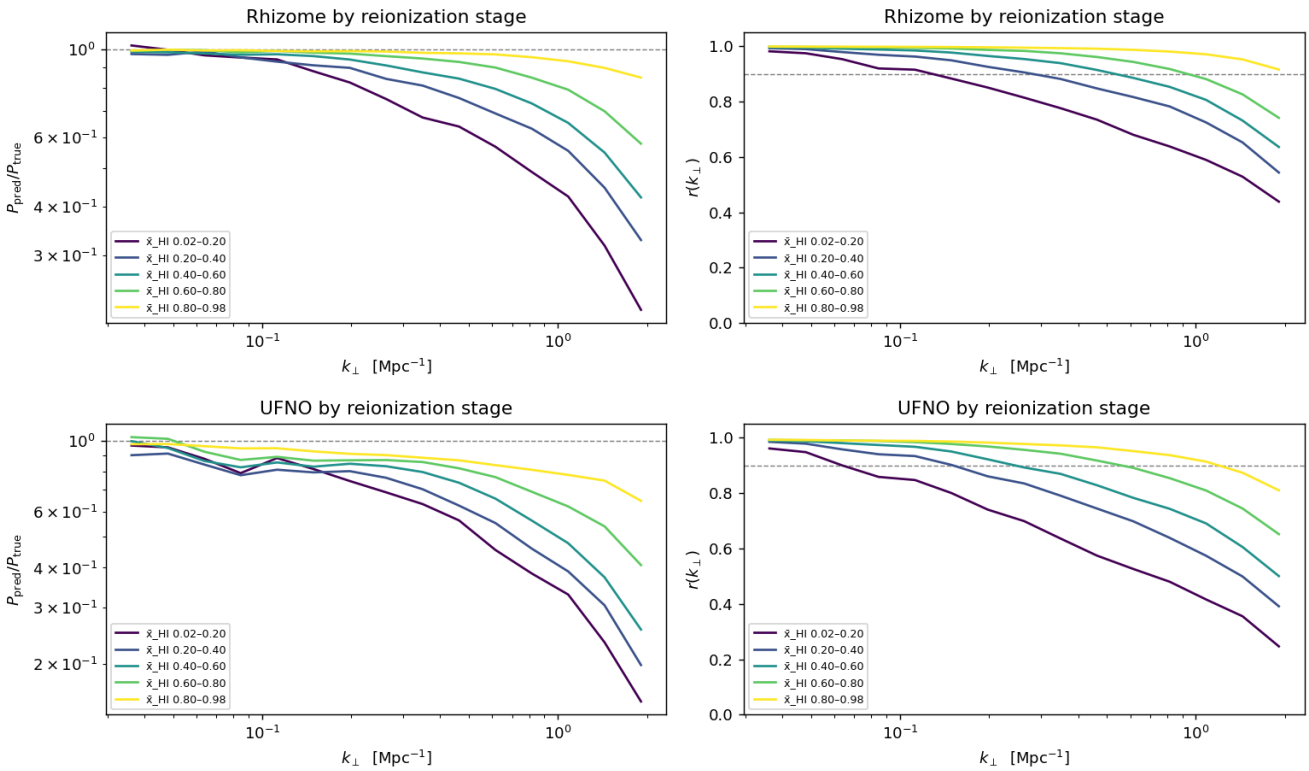


Figure 11. x_{HI} power ratio (left) and cross-correlation $r(k_\perp)$ (right) by reionization stage, multi-field slice-wise rhizome (top) and `fno` `tbs+histtb` (bottom), 199 test cones.

Table 12. Main results, 199 multi-field test cones (medians). "Slice-wise" is the multi-field slice-wise 2-D rhizome trained on all cones; "+ T_b power" adds the power term of Section 4.10. Best value per row in bold.

Metric	Slice-wise	+ T_b power	3-D rhizome C	`fno tbs+histtb`
$\lvert\Delta\tau_e\rvert/\sigma_{\rm Planck}$	0.010	**0.008**	0.012	0.076
$\lvert\Delta z(\bar{x}=0.5)\rvert$ / duration	**0.010 / 0.018**	**0.010 / 0.018**	0.011 / **0.018**	0.017 / 0.044
$x_{\rm HI}$ PDF W_1, active	**0.0095**	0.0098	0.0156	0.0271
b_1 pull	0.99	**0.93**	1.35	2.28
P_ε ratio (ideal 1)	**0.954**	–	0.945	0.793
bubble-size W_1, active (Mpc)	0.56	**0.48**	0.69	1.21
$x_{\rm HI}$ power error, high k	0.168	**0.163**	0.223	0.311
$1-r$, high k	**0.096**	0.102	0.151	0.174
edge H^1, ±2 Mpc	**0.106**	0.112	0.131	0.147
front width (Mpc; truth 2.77)	**3.89**	3.94	4.38	6.17
T_b PDF W_1, active (mK)	0.79	**0.77**	1.06	1.19
pre-reion. T_b power dev., $k=0.46$	$+24.8\sigma$	$\mathbf{+1.5\sigma}$	$+4.1\sigma$	$+12.8\sigma$

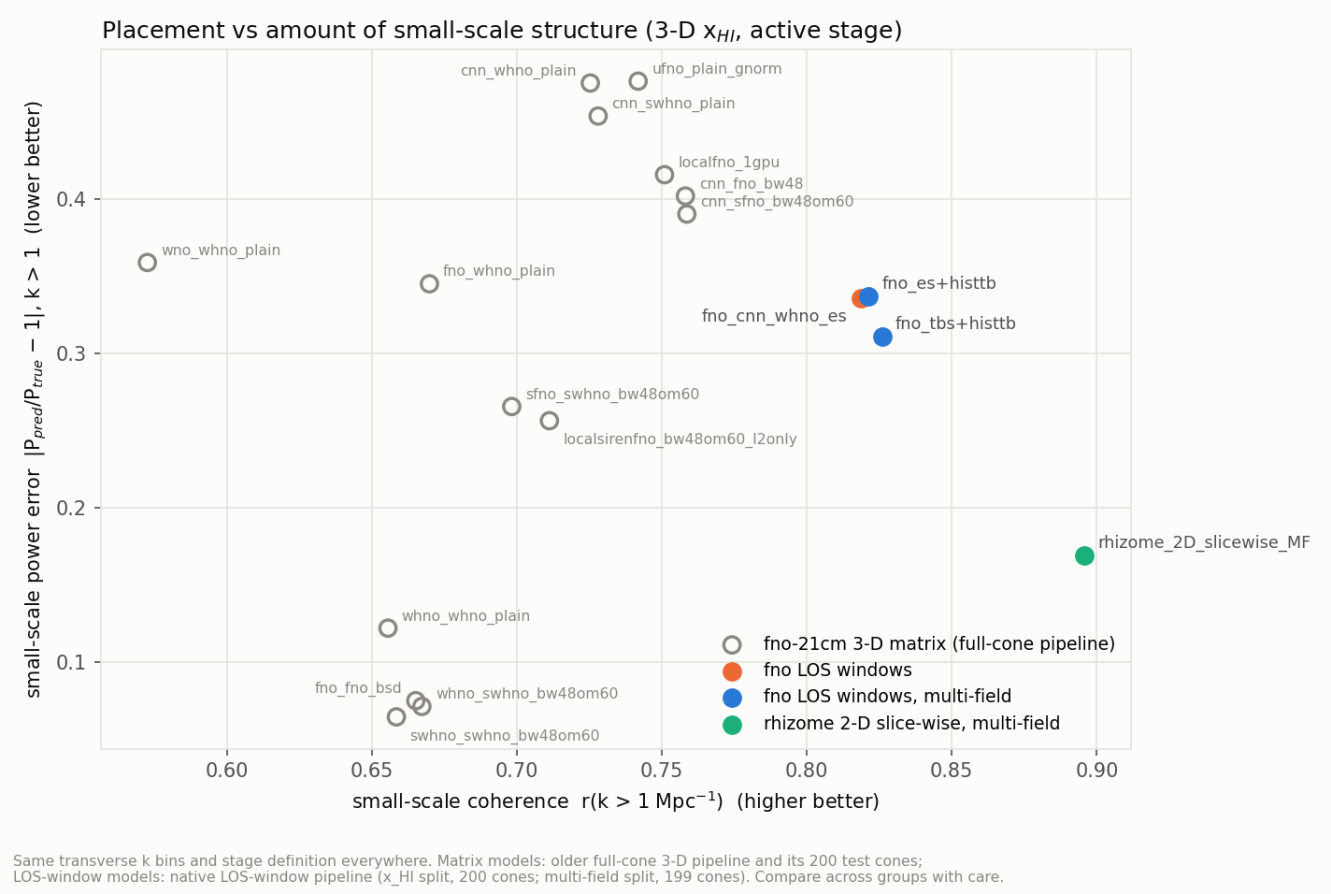


Figure 12. Small-scale coherence $r(k>1\,\mathrm{Mpc}^{-1})$ against small-scale power error for the 3-D $x_{\rm HI}$ models of the earlier operator study (open circles), the `fno` LOS-window models (filled) and the multi-field slice-wise rhizome (green), active stage. Better models lie to the lower right. Models from different pipelines and splits are compared with care.

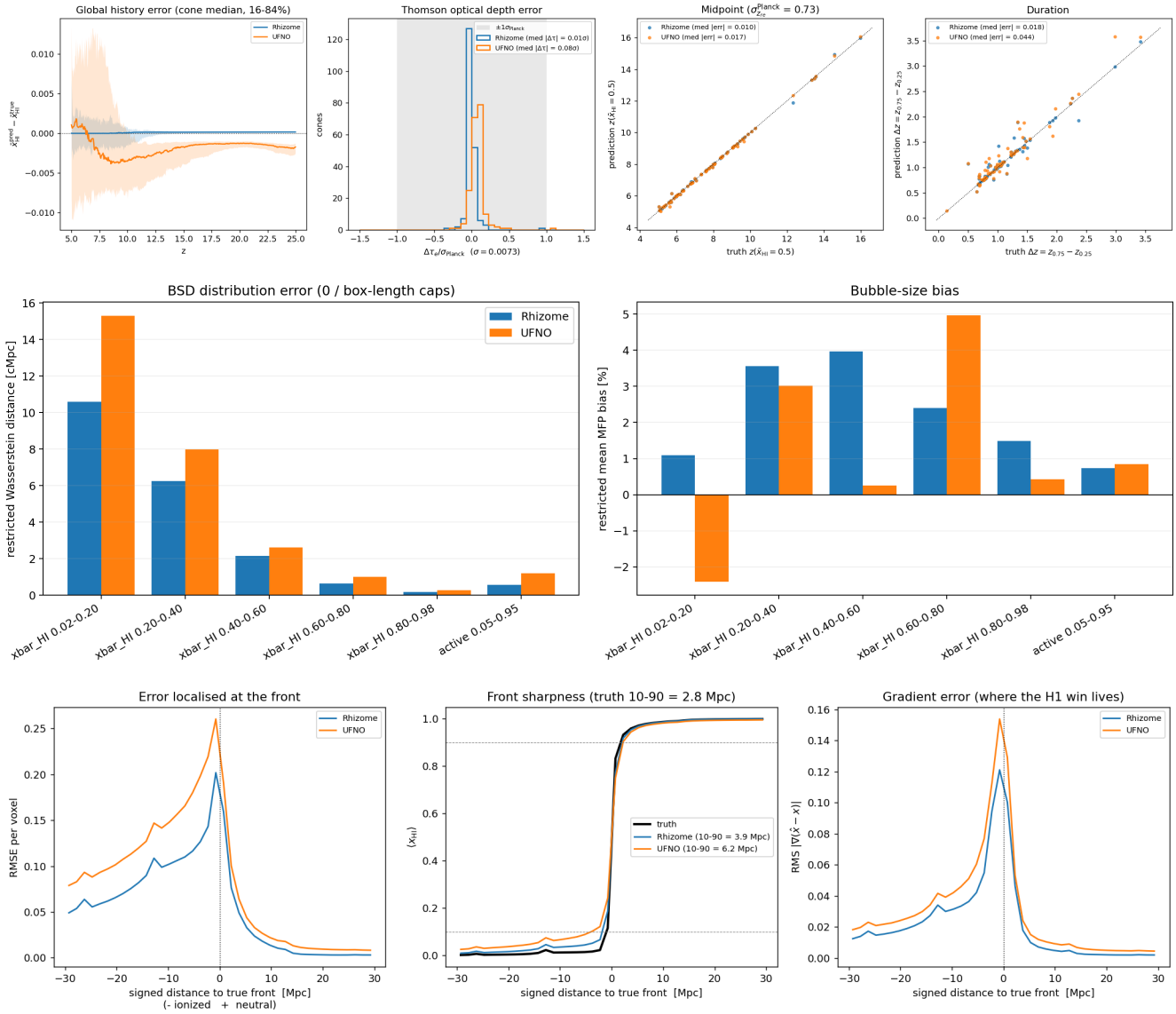


Figure 13. Physical metrics, rhizome (slice-wise multi-field) against `fno tbs+histtb`, 199 test cones. *Top:* global history error, τ_e error in Planck σ (median 0.01 against 0.08), and predicted against true midpoint and duration. *Middle:* bubble-size distributions and their W_1 distance to the truth by stage (active 0.56 against 1.21 Mpc). *Bottom:* error profiles across ionization fronts; front width 3.9 against 6.2 Mpc (truth 2.77).

Three fixes that address data, post-processing and context did not work; a fourth, in the loss, did (Table 13).

Table 13. Attempts to remove the absorption-era T_b power excess of the slice-wise model (power deviation at $k = 0.46\,\mathrm{Mpc}^{-1}$, plain model: $+24.8\sigma$).

Attempt	Excess	Outcome
3.9× more training cones	$+24.8\sigma$	T_b values improve (W_1 −31%), excess unchanged: no improvement from this data increase
smooth u along the LOS ($\sigma = 4$ cells)	$+1.1\sigma$	removes the excess only by destroying real LOS structure: T_b RMSE +40–110%, summed power ×1.3; rejected
LOS context ±128 cells	$+36.0\sigma$	worse; per-voxel accuracy unchanged; not adopted
small-scale T_b power loss	**$+1.5\sigma$**	reduces the reported excess across scales; costs 4–8% per-voxel RMSE

The power loss compares the transverse power spectrum of each predicted slice with the truth in eight logarithmic bins from $k = 0.3\,\mathrm{Mpc}^{-1}$ to the grid Nyquist frequency,

$$\mathcal{L}_P = \lambda \left\langle \left(\log(P_{\rm pred}+\epsilon) - \log(P_{\rm true}+\epsilon)\right)^2 \right\rangle_{\rm slices,\ bins}, \quad (25)$$

with $\lambda = 0.03$, applied to the mean-subtracted normalized T_b. It brings the absorption-era excess from $+24.8\sigma$ to $+1.5\sigma$, better than the 3-D rhizome, without overshooting the amplitude. It also improves the reionization-era statistics: T_b power within 1–2% at all scales, bubble-size W_1 −14% and τ_e −20% (Figure 15). The price is a 4–8% higher per-voxel RMSE, consistent over 80–90% of the cones, and slightly worse edges. This reflects the different quantities optimized by the two losses: squared error rewards agreement at each cell, while the power term rewards agreement in fluctuation amplitude. Matching a power spectrum does not guarantee correct phases or morphology, so cross-correlation and front metrics remain necessary.

Applying the same term to $x_{\rm HI}$ as well roughly halves the high-k $x_{\rm HI}$ power deficit (from 0.163 to 0.089 over active

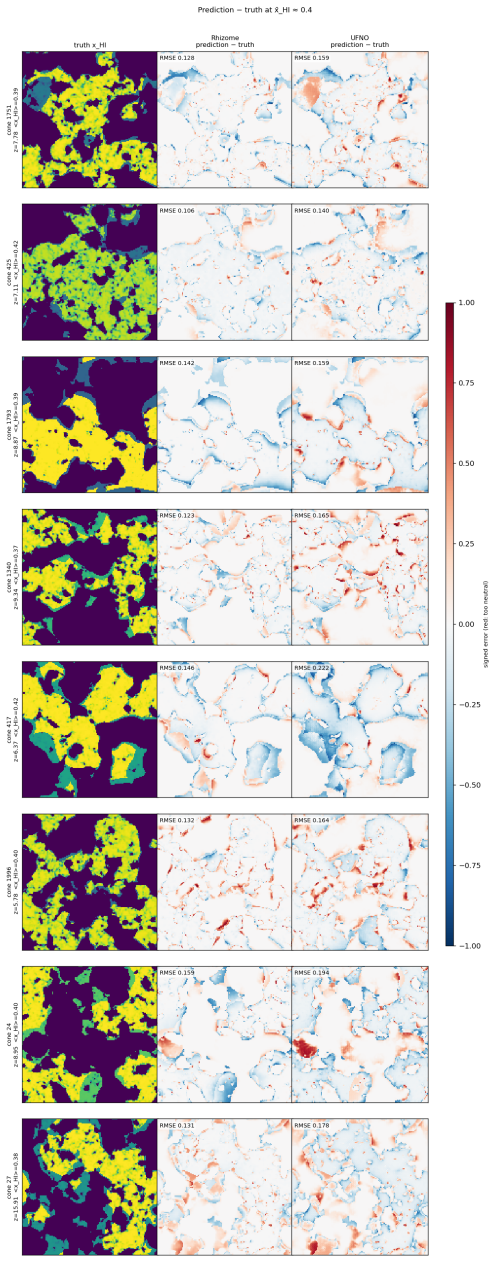


Figure 14. Signed $x_{\rm HI}$ error maps at $\bar{x}_{\rm HI} \approx 0.4$ on eight test cones (red: predicted too neutral), rhizome against `fno tbs+histtb`.

slices; in the latest stage from 0.59 to 0.15), gives the closest value distribution of all models (W_1 0.0082), and keeps the absorption-era fix ($+1.3\sigma$). It costs another 2–4% in per-voxel RMSE, and the extra small-scale power is only partly in phase with the truth (small-scale incoherence +7%). The three variants are useful for different purposes: the plain model for per-voxel accuracy, the T_b power loss for 21 cm statistics, and both power terms for closer small-scale power amplitudes in both fields, with the stated phase-error trade-off.

A further third of the absorption-era T_b error is a constant offset per slice, of random sign from cone to cone. Its size (1.6–2.0 mK RMS) is nearly the same for all three model families and comparable to the error of the global-history emulator that provides the $\bar{T}_b(z)$ input (about 2.1 mK). The similar error scales suggest that the history input may limit this component. Testing with true rather than emulated histories would separate that effect from model error. The power loss acts only on fluctuations and leaves the offset unchanged.

4.11 Learned spectral weights

Neural networks tend to learn low frequencies first, and a Fourier layer can end up using only its lowest modes. Figure 16 tests this for the rhizome by comparing the RMS weight of each Fourier mode of the trained kernel with its value at initialization. The weights grew 13–45-fold almost evenly up to the band edge: the ratio of high-quarter to low-quarter weight is 0.78–0.94 for all rhizome models, against 0.10–0.45 for a U-FNO in an earlier study. Models with 36 or 48 modes also develop non-negligible weights in their extra modes. Weight amplitude alone, however, does not measure mode usage: the transmitted input spectrum and output sensitivity also matter. These profiles show no collapse of the multiplier weights, but do not establish why the 48-mode sweep performed worse or rule out an alternative kernel parameterization. Longer matched-budget runs and per-mode message or sensitivity measurements would test those explanations.

4.12 Cost

Per training slice, the 2-D model is about four times slower than the 3-D models, since it runs full-resolution recurrent updates in float32. It needs 15–20 times fewer training slices, however (2.2 M against 33–49 M), so a complete training run is four to seven times cheaper: 3.5 h on one A100 for the best model, against about 18 h on an H200 for the 3-D rhizome and 44.7 h on an A100 for `cnn_whno_es`.

At inference, a full multi-field cone takes 5.5 s end to end on an A100 and 3.7 s on an H200 once the LOS bands are built on the GPU and the next data window is prefetched (Figure 17). This makes the slice-wise model the fastest model end to end, 1.8–2.3 times faster than both 3-D models; at that point inference is limited by reading the data. The GPU forward pass alone (3.8 s / 1.7 s) matches the 3-D rhizome and is about twice as slow as `fno`. Mixed precision is not worthwhile: bfloat16 gains only 5–14%, because the FFTs stay in float32, and it shifts $x_{\rm HI}$ by up to 0.08.

4.13 Ablations

Table 14 collects what each design and training choice was worth across the experiments above. The table includes controlled changes as well as comparisons of model configurations. All training runs use a single seed; the 2-D/3-D comparison changes several architectural and numerical choices together.

One planned ablation has not yet been run: a controlled study of the gate mechanism itself. It compares the full rhizome with six otherwise identical arms (no gates, source-only, receiver-only, gates computed from the conditioning only, gates computed once from the initial state, and static per-channel gates), five seeds each, all retrained from scratch. Until then, the comparison with `fno` establishes that the rhizome *as a whole* works better on this task, but not how much of the gain is due to the state-dependent gating specifically, as opposed to the recurrence, the loss and the LOS input.

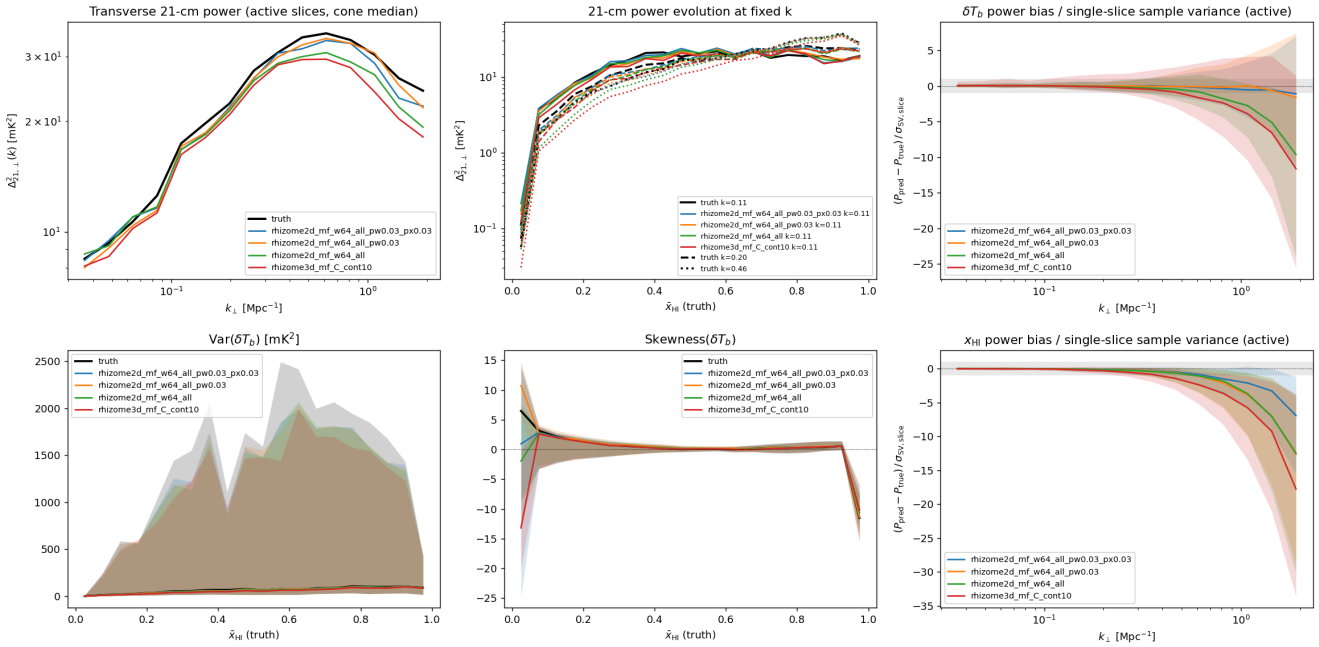


Figure 15. 21 cm statistics of the power-loss variants during reionization, 199 test cones: plain slice-wise model, + T_b power loss (`pw0.03`), + T_b and $x_{\rm HI}$ power loss (`pw0.03_px0.03`), and 3-D rhizome C. *Right column:* power bias in units of single-slice sample variance for T_b (top) and $x_{\rm HI}$ (bottom). Both power-loss models stay near zero to $k \approx 2\,\mathrm{Mpc}^{-1}$ in T_b; the plain model and 3-D C fall to -10 and -12σ.

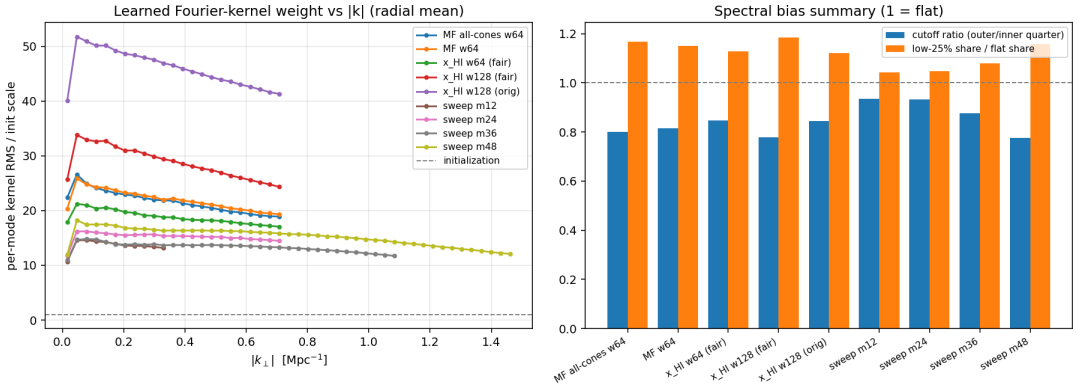


Figure 16. Per-mode RMS of the learned Fourier kernel relative to initialization. Radial profiles over wavenumber; they are flat to the band edge: there is no collapse onto low frequencies.

Table 14. Summary of ablations. Effects are relative changes in the stated metric; negative is better.

Change	Metric	Effect
LOS bands vs single slice (2-D)	mixed-slice RMSE	−36% (0.174 → 0.112)
width 64 → 128 (2-D)	mixed-slice RMSE	−3% at 4× parameters
width 48 → 64 (3-D)	test RMSE	+6% (worse)
lr $2\times10^{-3} \to 3\times10^{-4}$ (3-D)	test RMSE	+13% (worse)
warm restart (3-D)	test RMSE	−3 to −6%
history channels (3-D, A → B)	RMSE $x_{\rm HI}$ / T_b	−21% / −34%
structured T_b head (B → C)	T_b RMSE; timing	−5%; −15 to −30%
slice-wise 2-D vs 3-D (same data)	$1-r$; edge H^1	−40%; −21%
all 6,169 vs 1,594 training cones	median per-cone RMSE	−3.6% ($x_{\rm HI}$), −5.5% (T_b)
LOS context ±128 vs ±32	per-voxel RMSE	little reported change; duration −37%; absorption T_b worse
T_b power loss	pre-reion. T_b power dev.	$+24.8\sigma \to +1.5\sigma$; RMSE +4 to +8%
+ $x_{\rm HI}$ power loss	high-k $x_{\rm HI}$ power error	−45%; RMSE further +2 to +4%

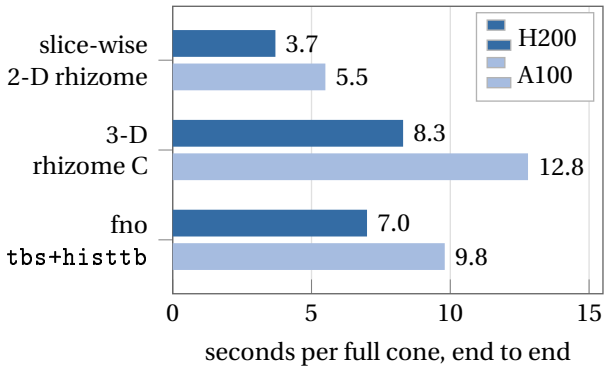


Figure 17. End-to-end inference time for one full multi-field test cone (x_{HI} and T_b), including data reading.

4.14 Discussion

What the rhizome does well. Across every configuration, the rhizome's advantage over the `fno` models is larger on physical observables than on per-voxel RMSE. It places ionization fronts more accurately (higher small-scale coherence), keeps them sharper, and reproduces bubble-size distributions, bias and stochasticity more faithfully. These are among the quantities that Section 1.5 identified as useful diagnostics of a restricted spectral interaction. They are also the ones that matter for 21 cm inference, where bubble morphology and the field-level bias enter the observables.

What the line of sight needs. The most consequential finding is not about the operator itself. A 2-D model with ±32 cells of LOS context, applied slice by slice, outperforms native 3-D models trained on the same data, and does so at a fraction of the cost. Together with the periodic tiling of the lightcones, this suggests that the bottleneck of the 3-D models is how they handle the LOS, not their transverse expressivity.

Scope of the evidence. The targets are semi-numerical lightcones from one simulation suite with repeated periodic boxes. Accuracy on those targets does not by itself establish agreement with resolved radiative transfer, unseen cosmologies, another box size or observational inference. The emulator is deterministic and has no calibrated field uncertainty. Power spectra, morphology diagnostics and optical-depth errors assess different aspects of agreement, but do not replace a downstream inference test.

Reproducibility details still needed. Before treating this report as a complete methods paper, the simulation-parameter ranges and sampling law, seed policy, split identifiers, input and target normalization, optimizer and weight decay, loss weights, history-emulator training split, and precise metric definitions should be recorded. The history inputs need an explicit provenance check showing that test-cone targets were not used to fit or select the emulator. LOS padding length must be distinguished from core and halo lengths. The meaning of the quoted σ and b_1 pulls, the partially ionized thresholds, and power-bin boundaries also need to be specified. These are missing documentation or checks, not assumptions supplied by this revision.

[**TODO:** Reconcile the 3-D model-C physical-suite numbers quoted in Section 4.7 with Table 12, or identify the differing checkpoint, subset or aggregation. Explain why the paired-bootstrap example uses 175 cones rather than the 200-cone test set, and specify its interval and resampling unit. Record whether the test comparisons informed the choice of context, loss weights or checkpoints.]

Caveats. Several confounds remain. All results are single seeds, and differences of a few percent (for example A against A0) may be partly noise. Part of the advantage in τ_e and the global history comes from training with BCE rather than from the operator, and the `fno` models should be retrained with BCE to separate the two. The contribution of the gating itself has not been isolated (Section 4.13). Finally, all results are at one resolution; the grid-independence of the parameters has not been put to the test.

A NOTATION

Symbol	Meaning
Operator learning	
f, g	input and output functions
$\mathscr{A}$, $\mathscr{U}$	input and output function spaces
D, N	spatial domain and total grid-cell count
$\kappa(r, r', \theta)$	kernel of an integral operator
$\mathscr{F}$, $\mathscr{F}^{-1}$	Fourier transform and inverse
$R_\phi(k)$	learned Fourier coefficients of the kernel (FNO)
$k_{\max}$, Λ	mode-count setting and retained index set
d_v	channel width
P, Q, W, σ	lifting, projection, pointwise linear path, nonlinearity
Rhizome operator	
$\boldsymbol{h}^t$, $\boldsymbol{c}$, $\boldsymbol{m}^t$	hidden state at update t, conditioning, message
A_θ, B_θ	receiver and source gates, values in $(0, 2)$
V_θ	value map
κ_θ, $R(k)$	base spatial kernel and its Fourier coefficients
κ_t^{eff}	effective, state-dependent kernel at update t
T, Δt	computational updates (6), step size (0.5)
21 cm physics and evaluation	
x_{HI}	neutral hydrogen fraction; $\bar{x}_{\mathrm{HI}}$ its global mean
T_b	21 cm brightness temperature (mK)
δ, z, ϑ	density contrast, redshift, simulation parameters
u	spin variable, $1 - e^{u}$ the spin-temperature factor
$T_0(z)$, F_v	T_b amplitude and velocity-gradient factor
τ_e	CMB optical depth to reionization
$r(k)$	cross-correlation coefficient of prediction and truth
b_1, P_ε	linear bias and stochastic power of the EFT expansion
W_1	Wasserstein-1 distance
LOS	line of sight

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